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AgriYieldAI an AI driven decision support system for crop yield prediction using YieldFusionNet

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Abstract
Crop yield prediction is a key issue that can contribute to food security by enabling better decision-making in agricultural planning, supporting food consumption, and reducing the effects of climate variability. However, the majority of current soil–vegetation modelling methodologies are either limited to single-source data or utilise linear statistical models, which do not capture the nonlinear interactions among soil health, time-varying atmospheric conditions, and fine-scale vegetation attributes derived from remote sensing. In recent years, Generic deep learning approaches have had shortcomings, such as the inability to generalise across regions, inefficiency, and low interpretability, which have discouraged their successful deployment in the field (on farms) as a workable tool in real-life agricultural settings. To fill these gaps, this paper presents AgriYieldAI, a decision-support system based on Artificial Intelligence (AI) for predicting crop yields using a newly developed deep learning method, YieldFusionNet, described in this work. Our architecture consists of three branches with three related modules (MLP, LSTM, and CNN) for soil data, time-series weather data, and satellite images (features from three sources). The model architecture is performance-, generalizability-, and explainability-centric—SHAP-based feature importance visualisation. SHAP-based analysis is applied to visualise feature importance based on NDVI, day5 rain, day30 rain, and soil pH from the day30 database. It was assessed on a range of real-world public datasets. As a result, it achieved an RMSE of 0.431, an MAE of 0.292, and an R^2 of 0.91, surpassing traditional state-of-the-art models such as Random Forest and Support Vector Regression (SVR). Additionally, a regional case study in the Guntur region of India illustrated the practical use of this method. AgriYieldAI may be a powerful, explainable crop yield forecasting solution that farmers, policymakers, and agronomic planners can use to obtain interpretable, data-driven insights for sustainable agricultural practices.

Keywords Crop yield prediction, Deep learning, Multimodal data fusion, Remote sensing, Explainable AI

1 Introduction

Agricultural productivity is a key driver of the world's food security, which will be put under pressure by a growing demand, exacerbated by climate change, extreme weather events, land degradation and resource scarcity (UN, 2020). Prediction of notable crop yields has become more analogous this time, in terms of speed and precision, so that resources are utilised at optimal levels, and policymakers in the supply chain make efficient decisions. Yield forecasting models usually rely on linear statistics or single-source data, which limits their transferability not only across regions but also under different conditions. Artificial intelligence (AI) methods, intensive learning, have recently demonstrated better performance in modelling complex linear and nonlinear relationships for multimodal data streams, such as soil properties, climatic conditions, and satellite images [1, 2].

Previous work on yield estimation relies on isolated modalities or black-box AI models that are not interpretable, difficult to generalise, or hard to deploy under resource constraints. Previous studies indicate that incorporating diverse data types, using models adaptable to regional contexts, and ensuring transparency in model interpretation are required [3, 4]. These gaps are a strong call for an AI framework that provides end-to-end, explainable, and scalable AI, translating into actionable insights for stakeholders at all levels.

The AgriYieldAI system is a sophisticated decision-support system for crop yield prediction that leverages on-demand (offline) and real-time (online) AI technologies. We hereby present the concept of a multi-mode, hybrid deep-learning framework, YieldFusionNet, which simulates the integration of indicators of soil health (SHI), climate (CI), and satellite-based vegetation indices. Our system can serve as a benchmark for generative reading and writing tasks and offers interpretability that surpasses earlier and SOTA methodologies. Our main contributions are a three-branch fusion structure that combines MLP, LSTM, and CNN pathways; the interpretability of our three-branch network via SHAP values; and a lightweight architecture with high accuracy for better deployability. We demonstrate a generic architecture, a building block validated on open datasets, an applied example for predicting regional yield, and an explainable model that provides critical features for agronomic decisions.

We organise the remainder of the paper as follows: Sect. 2 provides a review of related work on crop yield modelling and multimodal AI approaches. In Sect. 3, we explain the proposed methodology, including data preprocessing steps, model architecture details, training methods, and the explainability methods utilised. The fourth section presents experimental results, cross-dataset validation, model comparisons, and a regional case study. We present our key findings and limitations in Sect. 5 and conclude in Sect. 6, before discussing directions for future work to develop AgriYieldAI's capabilities and methods for deployment.

2 Related work

2.1 AI and machine learning techniques in crop yield prediction

AgricultureAI and machine learning (ML) are used in agriculture to analyse complex datasets and accurately predict crop yield. We have used models such as Random Forests, Support Vector Machines, and Deep neural networks to detect non-linear relationships of environmental variables with yield. Hybrid models that combine predictive algorithms have also been found to increase predictive performance elsewhere.

Han et al. [1] Effects of climate change, monitoring limitations, nature-based solutions, and pathways into the future. Mehedi et al. [2] Agriculture 4.0: Opportunities and Challenges in Decision Support Systems and the Role Of Remote. In particular [3], presented improvements to AI/ML-driven statistical models for disease prediction and discussed the obstacles to achieving stability in plant disease prediction systems, as well as the need for partnerships across the sector. Mana et al. [4] Farmers will be able to provide solutions to various problems in the farm sector, develop precision farming for future generations' needs, and help improve sustainable agriculture. Pandey and Mishra's [5] study discusses harnessing AI to achieve food security by reducing food waste and improving agricultural practices through cutting-edge technologies. Canata et al. [6] Combining artificial intelligence and multispectral data, scientists forecast sugarcane quality two months before harvest to help schedule harvest and make decisions.

Mohyuddin et al. [7] In their research, the authors state that machine learning (ML) is a key performance enabler for diverse concepts that transform the agricultural domain towards resource-efficiency optimisation, precision farming, and smart farming. A brief overview of recent research on irrigation scheduling using various data-based methods and models has been discussed in review articles by [8] and [9], including an ML approach using RS data and advanced algorithms to maximise crop water-use efficiency by determining optimal irrigation timing. In Martin et al. [10], a novel agricultural system relying on explainable AI was proposed to support sustainability and resource management, and to anticipate crop production.

Javaid et al. [11] Others reviewed the potential of AI for plant and soil health monitoring and for improving agricultural efficiency through the use of monitoring data and novel technologies. Araujo et al. In [12], ID3SAS, an AI-based system that makes data-driven decisions, was introduced to achieve optimal crop yield and water use in agricultural production. Saxena et al. [13] describe how artificial intelligence can assist with crop management, address agrarian challenges, and minimise the impacts of climate change on agriculture. Awais et al. [14] reported on sustainable agriculture and how AI, ML, and DL will help analyse the soil and overcome the disadvantages of traditional methods. Remote sensing methods for tracking soil degradation were reviewed by Wang et al. [15], who also identified challenges and opportunities for future sustainability research.

Gebresenbet et al. [16] proposed a framework for identifying impediments and outlining directions for future research in sustainable agriculture within the context of digital agricultural technologies. As stated in Javaid et al. [17], Agriculture 4.0 refers to the rise in food sustainability and productivity through digital solutions such as AI, IoT, and big data. Elbasi et al. [18] examined the use of artificial intelligence (AI) in agriculture, covering sustainable and efficiency improvements enabled by robotics, the Internet of Things, and machine learning. A review of AI and IoT, as reported in Raj et al. [19], also covers their implementation in precision farming and proposes approaches that enable growth maximisation, yield, and profit. An overview of the applications and challenges of Ag-IoT systems for agriculture has been given by Chamara et al. [20]. It could be the rigid commercial platforms. Future research opportunities lie in machine learning, mobile IoT, and satellite-based connectivity.

Sarma et al. [21]: An IoT- and AI-based innovative agricultural system for disease monitoring and detection using CNN. One drawback is that the approach depends on specific conditions. Future research may look into additional crops and climates. Jeong et al. [22] developed a hybrid model to predict yield at the pixel level—one of its shortcomings is that it depends on the region. Further studies may include larger applications. Gonzalez et al. [23] analysed machine learning (ML) methods for soil quality assessment using remote sensing (RS) data, identified the challenges, and proposed a model for local-scale implementation. Key limitations Include Medium- and low-scale systems. Enhancing regional models is a promising avenue for future work. Kganyago et al. [24] reviewed machine learning (ML) and remote sensing for biophysical and biochemical estimation, discussed relevant problems, and presented semi-scalable models for poorly characterised regions. Feng et al. [25]. This review summarises the challenges in genomic prediction, evaluates the impact of AI on crop breeding, and identifies possible directions for future research to achieve sustainable agriculture.

In another study of wheat cultivation, Gawdiya et al. [26] developed ensemble machine learning models that predicted wheat production from morpho-physiological characteristics with higher accuracy. Future work will involve fine-tuning advanced models. In a recent study led by Shi et al. [27], forecasting models were developed to assess the agricultural bioenergy potential of the Plain, with particular focus on the impacts of temperature. Future work will focus on optimising regional climate plans. Santos et al. [28] evaluated random forests and linear regression to determine the optimal hyperparameters and data thresholds for predicting soybean yield. Asadollah et al. [29] describe the RScv optimisation, designed to improve prediction accuracy and increase the global transferability of predictions by combining machine learning methods used in crop production prediction across Europe. Mitra et al. [30] predicted cotton yields using machine learning, incorporating management, soil, and climate components to support climate-smart agriculture.

Islam et al. [31]. Then, a cotton leaf disease detection system using optimised deep learning models was proposed, achieving 98.7% accuracy. Eneh et al. [32] reported a transportable aquaponics IoT water-quality sensor system with a 99% accuracy in predicting fish growth. In their solution, which achieved 98.66% precision in potato leaf disease identification, Arshad et al. [33] presented a hybrid deep learning network called PLDPNet. The online application was developed in [34] with an accuracy of 98.98% for the early detection of crop diseases using ResNet-50. The accuracies and inaccuracies of the GEP, ANN, and REG models for predicting rice growth rate were examined by [35].

Jung et al. [36] demonstrated how advances in AI and remote sensing could ultimately enhance the resilience of agricultural systems and address future challenges to food security. The Deep Recurrent Q-Network, a model that integrates Q-learning with deep learning, was introduced by Elavarasan and Vincent [37] to predict crop yield with 93.7% accuracy. Murali et al. [38] developed WOARNN-GARCH, a hybrid model for forecasting sugarcane yield with volatility, combining GARCH, RNN, and Whale Optimisation. Rashid et al. [39] reviewed machine learning algorithms for forecasting agricultural yield, focusing on palm oil, identified challenges, and proposed future work. Ma et al. [40] ADANN is an unsupervised, adaptive domain-adversarial neural network that demonstrates superior corn yield prediction across regions compared to existing models.

Future work can focus on implementing modifications to improve the model's transferability and applicability to specific crop types.

Kaur et al. [41] 's CropYieldNet refers to work on developing a multimodal-based early prediction of crop output; an overview of the system is presented in Fig. 1. In this work, we aim to enhance the model's generalisation. Nayak et al. [42] addressed variability in wheat yield by comparing machine learning methods and found that random forests were the most applicable. Next steps: explore additional production methods. Aworka et al. [43] developed machine-learning-based crop prediction models in East Africa to estimate future production. Higher data availability can be factored into future work. Cédric et al. [44] developed machine learning algorithms to predict crop yields in West Africa. Future research may primarily focus on enhancing the model's generalisation and accuracy. Ji et al. [45] MODIS data enabled the development of a generalised model for crop yield prediction that demonstrated good performance but was less precise.

Singla and Bali [46] investigated hybrid and machine learning models for crop yield prediction, accounting for climate and deep learning [45]. applied XAI to maize yield data to enhance the transparency of cultivation forecasts [46]. Moreover, Goel and Mishra [47] reviewed agricultural yield prediction systems, reporting that deep learning is most accurate and that methods perform best for a specific crop. Luo et al. Precise mapping of sap and prediction of wheat production are essential for global food security research. Priyatikanto et al. [48], Domain-adaptable machine learning models for maize yield prediction, as demonstrated, can be broadly applicable across various geographical regions.

Sabo et al. [49] examined the literature on deep learning models for predicting crop yield. They demonstrated that, when comparing the predictive performance of machine learning and deep learning models, deep learning models are superior. Jhajharia et al. [50] employed machine-learning techniques to predict yield and found that Random

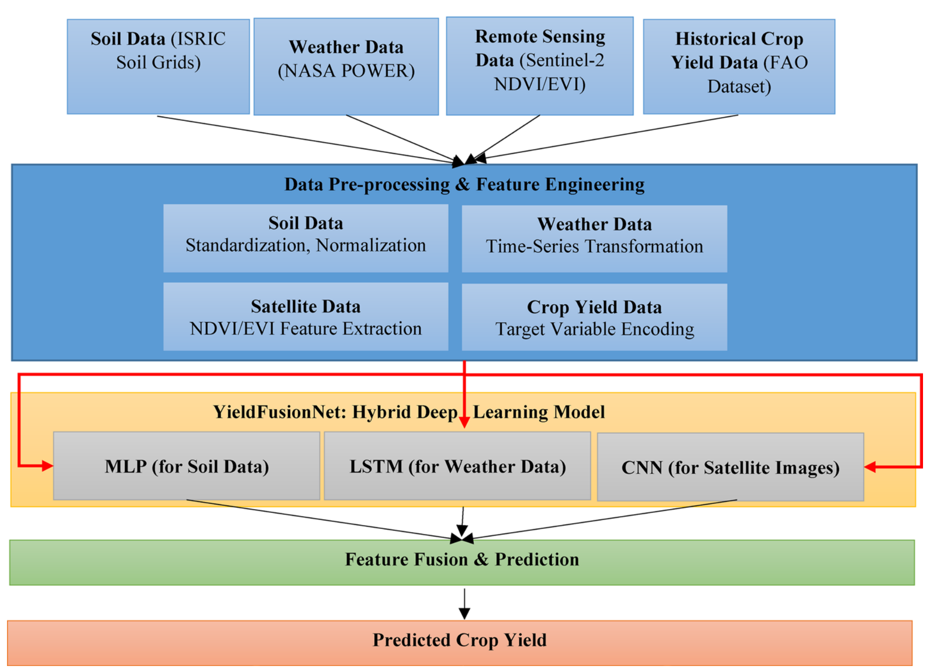

Fig. 1 AgriYieldAI – An AI-Driven decision support system for crop yield prediction integrating soil, weather, and remote sensing data

Forest achieved the highest accuracy. Chuan et al. Jiménez et al. [51] presented a stacking ensemble learning model based on support vector regression (SVR), random forests (RF), and K-nearest neighbours to robustly and accurately predict soybean yield. Hai et al. [52] Machine learning prediction of SSA and biochar yield from 46 biomass types; Random Forest is the best (R2 = 85%). Stiller et al. [53] To illustrate, a paper that compared RCV and SCV for deep learning-based crop yield prediction showed that SCV yielded better transferability than RCV in smallholder agriculture. (more context) Future work could explore additional data and scalability.

Kartick et al. [54] A streamflow prediction model based on machine learning utilising significant predictors and random forest. It might also be utilised internationally in future investigations. Andreu et al. [55] propose a YOLOv8-based generic tool for real-time crop pest prediction, which is part of future work, as outlined in Jeong et al. [56] Rice (AnywayLSTM-LSTM). Based on deep learning and remote sensing to predict rice yield, LSTM is the most effective. Focusing on building diverse training datasets remains as future work. Ashfaq et al. Random Forest provides better predictions than other models for wheat yield across seasons, with models developed by [57] integrating climate, satellite, soil, and socio-economic data. Tunio et al. [58] propose a system that estimates agricultural yield, guided by meta-knowledge, hyperparameter optimisation, and model transferability across datasets.

In evaluating machine learning models for predicting sweet corn production, the study by Dhaliwal and Williams [59] found that year, location, and genetics were essential factors. You to Dimensionality reduction and Deep learning for crop yield estimation using remote sensing: A survey of South India with 98.96% accuracy [60]. Hoque et al. [61] A prediction system for crop production was developed from climate, pesticide, and crop yield data, achieving 99.99% accuracy with the Gradient Boosting technique. A deep neural network framework was developed by Demirhan [62] to forecast the future production of principal crops under climate change scenarios in Australia, outperforming several benchmark models for crop yield prediction in the country. Salam et al. [63] developed a more accurate crop yield forecast using an improved SVR model and a hybrid feature selection method. Bloh et al. [64] For example, improved wheat yield predictions by transferring knowledge from DSSAT crop simulations into process-based and machine learning models.

Recent studies have applied machine learning, deep learning, and remote sensing techniques to improve agricultural monitoring, crop prediction, and climate adaptation. Schulz C et al. [65] used Sentinel-2 satellite time-series data and machine learning to monitor winter catch crops, while X Qin et al. [66] analysed factors influencing rural economic development in agricultural villages. A. Elbeltagi et al. [67] examined the impacts of climate change on the water footprints of wheat and maize production. J Brinkhoff and A J Robson [68] proposed spatio-temporal models for macadamia yield forecasting, and N R Prasad et al. [69] applied Random Forest for cotton yield prediction. Climate-based crop simulation and adaptation strategies were explored by P N Kephe et al. [70] and M D M Kadiyala et al. [71]. Deep learning applications in agriculture include CNN-based tomato disease detection by M Agarwal et al. [72], crop classification using remote sensing by D. M. Johnson and R Mueller [73], recurrent convolutional networks for crop recognition by J. A. Chamorro Martinez et al. [74], and weed detection models by A Ahmad et al. [75]. Additionally, S Traore et al. [76] developed the YIELDCAST tool for rice yield forecasting under climate variability. At the same time, N Kumar et al. [77] proposed a framework for assessing the risk of climate-resilient infrastructure. These studies collectively highlight the growing role of AI, remote sensing, and climate modelling in improving agricultural productivity and sustainability.

2.2 Integration of remote sensing and IoT in precision agriculture

Remote sensing technologies have significant potential for data acquisition and monitoring in agriculture when coupled with IoT. Data from satellite images, UAVs, and ground sensors provide real-time information on crop health, soil conditions, and environmental factors, allowing precise interventions and optimal resource use.

Han et al. [1] This news alerts us to the hotspots we need to track, spot trends and respond to, while reaffirming that nature-based solutions and the solutions we build on their back truly matter in the face of climate change — and the ongoing impacts on ecosystems. Mehedi et al. [2] Agriculture 4.0: Potential, challenges, and role of remote sensing. In [3, 5], the writer has outlined achievements in AI/ML-based disease prediction models, challenges in the agricultural sector, and the need for collaboration.

Canata et al. [6] To leverage harvest scheduling and decision-making, we investigated the feasibility of forecasting sugarcane quality using artificial intelligence (AI) and multispectral data. Munaganuri and Rao [12] proposed a machine learning model that combined remote sensing and advanced algorithmic tools to improve irrigation scheduling and more effectively utilise essential water supply for crops. Araujo et al. In particular, developed an intelligent decision-support system (ID3SAS) grounded in an artificial-intelligence-based, data-driven approach to maximise agricultural yield and water utilisation.

Awais et al. [14], from traditional methods that hinder sustainable agriculture, to AI, ML, and DL services to enhance soil analysis. Wang et al. [15]. Along these lines, Bhowmik et al. [20] systematically reviewed remote sensing methods for soil degradation assessment and similarly provided an overview of related indicators, constraints, and future research directions towards sustainability. Chamara et al. Jahanshahi et al. According to Ag-IoT systems, they utilise various agricultural applications but also present some challenges. A disadvantage here is the inflexibility of commercial platforms. In the future, our R&D will focus on robotics, machine learning, and satellite and mobile IoT connections.

Sarma et al. [21] proposed an innovative agriculture system for disease monitoring and detection using IoT and AI with CNN. They have several limitations, such as Dependence on specific conditions. Future research may explore other crops and climates. Jeong et al. [56] Oil-palm pixels could serve as a proxy for hyperspectral-scanned rice pixels, and [22] hybridised models to estimate rice yield at the pixel level. There are some drawbacks, including regional dependence. Future studies may include broader applications. Gonzalez et al. [23] reviewed several issues related to machine learning (ML) methods for soil quality assessment using remote sensing (RS) data and local-scale models. Medium- and low-scale systems are not limited by these factors. This, however, might clear the way for improved regional models in the future. Kganyago et al. [24] studied ML and remote sensing for biophysical and biochemical estimates, identifying problems and proposing scalable models for poorly characterised regions.

Asadollah et al. [29] RScv for better accuracy and global transferability of crop production prediction across Europe using machine learning methods. Ji et al. [45] A generalised model for crop yield prediction using MODIS data was created and, though reasonably well-performing, has room for further development.

Luo et al. [78] developed the GWPMS framework for mapping wheat production areas and yields, thereby improving the accuracy of research on global food security. Domain-adaptable machine learning models have been applied to maize yield prediction across various geographical regions [48]. Stiller et al. [53], by comparing SCV and RCV for deep learning-based crop yield prediction, showed that SCV has better transferability in smallholder agriculture. Future work can explore additional data and scalability.

Jeong et al. [56]. LSTM outperformed other classical deep learning techniques for predicting rice yield when fused with remote sensing data. Future work should focus on more diverse training datasets: Ashfaq et al. [57], Machine Learning Models for Predicting Wheat Yield: A Data Fusion Approach. Combine climate, satellite, soil, and socioeconomic data to predict yield and outperform other methods using Random Forest. Hoque et al. [61] used weather, pesticide, and crop yield data to build a 99.99% accurate crop production prediction system using Gradient Boosting.

2.3 Decision support systems and explainable AI in agriculture

AI-based Decision Support Systems (DSS) deliver actionable insights to farmers on crop management. Explainable AI (XAI) for transparency in AI-based recommendations to build trust and informed decision-making in agriculture.

Mana et al. [4] They elaborated on the applicability of AI to address challenges in harvesting, improve precision farming for future needs, and support sustainable agricultural practices. Looking at a wider picture, Pandey and Mishra [5] reviewed the use of artificial intelligence to maintain food security, reduce food waste, and optimise agriculture using advanced technologies. Martin et al. In this work, we described an innovative agricultural system that uses AI to enhance sustainability, improve resource management, and enable efficient crop yield prediction, all grounded in XAI [10].

Araujo et al. [12], ID3SAS involves multiple components of spatiotemporal decision-making. An AI-enabled spatiotemporal agriculture decision-making system that monitors the cultivation process, ensuring every collected data point is highly relevant and capable of increasing crop yield and reducing water use. Saxena et al. [13] Aggressive use of AI in crop management & agricultural challenges [14] and Climate change impact on agriculture. Gebresenbet et al. [16]. Due to the disciplines it spans and its set of indicators, this model could serve as a framework for conceptualising obstacles and informing future research on how digital technologies can advance sustainability in agriculture. Javaid et al. [17] Title of the image: Agriculture 4.0 — What is Agriculture 4.0. Leveraging digital tools, including Artificial Intelligence (AI), Internet of Things (IoT), and Big Data, to enhance the sustainability and efficiency of agriculture. Elbasi et al. [18] provide a review of AI applications in agriculture, outlining methodologies that use Robotics, IoT, and Machine Learning to enhance sustainability and productivity.

The paper uses a model to estimate production in East Africa. Another limitation is that future studies can address the issue of improving data availability. Cédric et al. [43] Yield predictions for West African crops were made by [44] using machine learning algorithms—future research may focus on improving the model's generalisation and accuracy. Recent studies that display comparable features include Ryo [79], in which maize yield data were subjected to XAI to improve transparency in agricultural predictions. In the following work, Goel and Mishra [47] studied several farm yield forecasting systems and noted that deep learning methods and crop-specific approaches achieved high accuracy in yield prediction.

Priyatikanto et al. [48] Investigation on domain-adaptable machine learning models to predict maize yield and their universality to various geographical regions. Stiller et al. [53]. The transferability of SCV, as opposed to RCV, for deep learning-based crop yield prediction was demonstrated, as it is more favourable in smallholder agriculture contexts. Future research could explore more data and scalability.

LSTM is the most effective method and has been adopted to predict rice yield by integrating deep learning (DL) and remote sensing [56]. In future research, greater emphasis should be placed on training datasets with greater diversity. Random Forest yields the best results among the models in predicting wheat yield by integrating climate, satellite, soil, and socio-economic data [57]. Hoque et al. [61] Using weather, pesticide, and crop yield data, developed a crop production prediction system that achieved 99.99% accuracy using Gradient Boosting.

2.4 Climate change impacts on agriculture and adaptive strategies

Climate change poses significant challenges to agriculture, crop yields, and food security. After this Wide-Ranging Research on climate impacts in agriculture, many researchers developed adaptive strategies to mitigate the effects of climate change and promote sustainable agricultural practices.

Climate change is wreaking havoc on ecosystems, but we cannot fully observe its impact on our natural world. We need nature-based solutions and more projects to combat climate change. New Approach to AI/ML-based Disease Forecasting: Recent advancements in the field of AI/ML-driven disease prediction models, challenges, combating the common challenges in the agricultural sector, and the importance of collaboration [3]. Saxena et al. [13]. AI has potential applications in crop management, addressing agrarian problems, and mitigating the effects of climate change on agriculture.

Shi et al. [27] confirmed earlier research and developed forecasting models for agricultural bioenergy potential in the Plain, with a focus on temperature effects. Subsequent research will involve enhancing regional climate strategies. Andreu et al. [55] proposed a YOLOv8-based tool that is broadly zoo-chained and achieves effective results for international crop pest detection. Expansion of the dataset is among the future work. Jeong et al. [56] predicted rice yield using deep learning and remote sensing, achieving the best performance with LSTM. Future work should focus on diverse training datasets.

Ashfaq et al. [57] By integrating climate, satellite, soil, and socioeconomic data, Martin et al. [59] predicted the wheat yield using machine learning models, with Random Forest outperforming other approaches. Demirhan [62] developed a deep neural network framework for predicting future crop yields under climate change scenarios, significantly outperforming benchmark approaches in Australia. Bloh et al. Improving wheat yield forecasting via knowledge transfer from crop models in DSSAT. This is a joint contribution by the authors listed alphabetically here.

According to Kephe et al. [70], the lack of data is a fundamental barrier to successful crop modelling in South Africa, and they propose a hybrid approach to data procurement. In India, the effects of climate change on groundnut production are highly variable, and Kadiyala et al. [71] here explored the biophysical and socio-economic impacts. Ojo and Baiyegunhi [80] studied the effects of climate change adaptation on rice output in Nigeria, including methods and outputs.

3 Proposed system

In this work, we propose and develop AgriYieldAI, a novel AI-based framework for crop yield prediction that leverages multimodal data sources, covering data collection, processing, modelling, and accuracy assessment. We propose a hybrid deep learning architecture—YieldFusionNet—that leverages synergistic interactions among soil health parameters, time-series weather sequences, and remotely sensed vegetation indices to enhance prediction accuracy, generalizability, and explainability across diverse agricultural regions.

3.1 Overview of AgriYieldAI framework

AgriYieldAI is an AI-based intelligent Decision Support System that integrates and interprets various data modalities (listed below) (Soil health data metrics, Weather data, remote-sensor images, etc.) to deliver accurate yield predictions (Fig. 1). Existing crop yield estimation methods rely on separate data, leading to inconsistent yield predictions across different agricultural zones. The AgriYieldAI system overcomes these limitations through a multimodal fusion approach, delivering credible, actionable, data-driven insights to farmers, agronomists, and policymakers. The framework is made generalizable using machine learning, deep learning, and feature engineering, enabling its application across a wide range of climatic and soil conditions.

The system integrates publicly available data sources at the mid-scale level, including soil health parameters from ISRIC SoilGrids, weather variables from NASA POWER Climate Data, vegetation indices (NDVI, EVI) derived from Sentinel-2 satellite images, and historical crop yield data freely available from the FAO Crop Production Statistics. Soil data provides information on nutrient composition, weather data shows fluctuations in each climate variable, and remote sensing imagery reveals plant health, all of which are essential components of crop yield. AgriYieldAI employs robust data preprocessing methods (feature normalisation, time-series transformation, and remote-sensing data augmentation) to extract generalisable patterns from heterogeneous datasets.

AgriYieldAI is built on a first-of-its-kind hybrid deep learning model– the YieldFusionNet, which can combine both structured and unstructured data to provide higher accuracy yield predictions. The model has three parallel learning paths: a Multilayer Perception (MLP) for soil data, a Long Short-Term Memory (LSTM) network for weather time-series data, and a Convolutional Neural Network (CNN) for extracting spatial features (from satellite images). From each independent processing stream, domain-specific features are extracted and passed through a feature-fusion layer to produce a final crop-yield prediction. The proposed new system architecture relies on LSTM neural networks to exploit both short- and long-term dependencies underlying crop growth, making it more robust than conventional statistical or regression-based methods.

This enhances the precision and scalability of prediction while maintaining a lightweight computational framework, conducive to implementation in high- and low-resource contexts for real-time agricultural decision-making. Another feature: AgriYieldAI also has an explainable AI that explains the rationale for the model's predictions (e.g., via SHAP (Shapley Additive exPlanations)). This may be key to building trust and adoption among agricultural actors because end-users will be aware of the factors driving the anticipated yield values. Built on a scalable platform, AgriYieldAI can be more readily integrated with an IoT-based agrarian monitoring system and a cloud-based large-scale platform. Such research is expected to promote model applications that are generalizable across agro-climatic zones to advance precision agriculture, climate resilience, and sustainable agriculture. Table 1 highlights the notations for the input features and the representations of intermediate and output variables used in the proposed methodology.

While it does predictive modelling, AgriYieldAI is built to be a holistic Decision Support System or DSS that translates yield predictions into actionable agricultural insights. The system incorporates a well-defined workflow that includes the collection of multimodal observation records, predictive modelling using YieldFusionNet, and post-predictive analytics processing to assist agronomic decision-making. Crop forecast estimates are plotted alongside soil health indices, climate risk factors, and veg state to enhance stakeholders' situational awareness for crop planning, resource allocation, and

Table 1 Notations used in the AgriYieldAI framework and YieldFusionNet model

Symbol	Description
S	Soil feature set (pH, nutrients, moisture, organic carbon)
S'	Transformed soil feature embedding after MLP
W	Weather data feature set (temperature, precipitation, humidity, solar radiation)
W_t	Weather data at time step t
$W_j^{(t)}$	Time-series weather sequence for feature j
h_t	LSTM hidden state at time step t
h_T	Final hidden state encoding climate impact.
c_t	LSTM cell state at time step t
i_t, f_t, o_t	LSTM input, forget, and output gates at time t
F	Remote sensing feature set (NDVI, EVI) from Sentinel-2 images
F_{l+1}	Feature map at CNN layer $l+1$
W_c	CNN convolutional filter weights
b_c	CNN convolutional bias
F_{CNN}	Extracted spatial features from CNN
F_{fusion}	Concatenated feature representation from soil, weather, and satellite data
W_F, b_F	Weights and bias of final dense layer
$\widehat{y}$	Predicted crop yield
y_i	Actual crop yield for sample i
θ	Model parameters at iteration t
η	Learning rate for model optimization
W_q	Quantized model weights for edge deployment
D_t	Preprocessed dataset at time t
$D_{raw}\left(t\right)$	Raw incoming agricultural data at time t
$\varphi_{\ i}$	SHAP feature importance value for feature i
I_j	Permutation importance score for feature j
$W^{(t)}$	Model parameters in federated learning at time t
n_i	Number of training samples in local dataset of client i
$\mathcal{L}_i\left(W^{(t)}\right)$	Local training loss function for federated learning client i

risk mitigation. Such a DSS-oriented design allows AgriYieldAI to go beyond forecasting by providing numerical results and instead supports real agricultural decision-making processes.

Typical universality with cross-crop transferability is not the goal for AgriYieldAI; instead, its generalisation capability is evaluated based on regional portability. The proposed framework can be adaptable across agro-climatic regions by utilising global soil, climate, and remote sensing datasets, with uniform data representations. Although the present study is based on homogeneous crop type-region settings, the framework can be extended to heterogeneous crop types, which is identified as one of the future research directions.

3.2 Data sources and preprocessing

The yield prediction system for AgriYieldAI is generalizable and scalable across regions, using soil health metrics, weather variables, and remote-sensing imagery (satellite images). It is constructed on public datasets to allow the system to be applied across different agricultural areas, soil compositions, and climate conditions. As ground-truth labels to train our models, we use historical crop yield observations from the FAO Crop Production Dataset [81] at the global scale. Soil fertility indicators (pH, organic carbon, nitrogen, and moisture contents) were fetched from the ISRIC SoilGrids database [82] at a coarse spatial resolution. NASA POWER Climate Data [83] provides globally consistent climate records from which we can retrieve various weather attributes, including temperature, precipitation, humidity, and solar radiation. Furthermore, vegetation indices (VIs) such as the Normalised Difference Vegetation Index (NDVI) [82] and the Enhanced Vegetation Index [83] are available as remote-sensing-based ground indicators from Sentinel-2 satellite imagery [84], serving as proxy data for vegetation biomass.

AgriYieldAI multimodal datasets have heterogeneous or complementary spatial and temporal resolutions, so they require explicit reconciliation before fusion. Sentinel-2 high-resolution remote sensing imagery is pooled to the region level to match coarse soil grids and administrative-scale yield records. To preserve spatial representativeness while avoiding pixel-level leakage, vegetation indices (NDVI, EVI) are summarised for each target region using mean and variance statistics.

Daily weather observations are temporally aggregated using rolling-window statistics based on crop phenology cycles, while satellite observations are aligned to the nearest valid acquisition window. Timestamps that are missing or asynchronous are processed using linear interpolation and window-based smoothing to enable consistent time indexing across modalities. This strategy for aligning modalities enables consistent multimodal fusion whilst retaining dominant agronomic signals from each data source.

These datasets are collected from multiple sources and have different temporal and spatial resolutions; thus, preprocessing is often performed to homogenise them and improve model performance. Since the soil data is mainly presented as spatial grids, it is interpolated across different depths and then normalised into tidy tabular representations. Each soil feature S_i is normalized as in Eq. 1.

$$S_i^{norm} = \frac{S_i - \mu_S}{\sigma_S} \ \forall \ \ i = 1,2,\ldots n \tag{1}$$

where S_i^{norm} is the soil feature, μ_S and σ_S and is the mean and standard deviation of the soil parameter globally for soil regions. Weather data, by nature, is temporal and needs to be converted tinto well-functioning slices to bereliable for deep learning models. This results in multiple input sequences for the LSTM model for each climate variable W_j, by using a rolling window approach over a set duration T as in Eq. 2.

$$W_j^{(t)} = \left[W_j^{(t-T)}, W_j^{(t-T+1)}, \ldots, W_j^t\right] \tag{2}$$

where $W_j^{(t)}$ is the climate variable at time t and the window size T is selected according to agricultural cycles seasonal patterns. To preserve temporal continuity, any missing values in the weather data are linearly interpolated.

So-called agricultural weather and satellite time-series data typically exhibit irregular sampling intervals as a result of missing observations; sensors outages, or atmospheric interference (e.g. cloud cover). All the weather variables are reindexed to a common time grid before they are sequentially modelled. Linear interpolation fills short gaps in missing values, and longer gaps are smoothed by means of rolling-window aggregation in-time domains with crop growth cycles. These fixed-length sequences are then sliced through sliding windows and passed as input to the LSTM model. This just-in-time preprocessing allows the RNN to still learn seasonal and phenological patterns, but is temporally consistent across samples. Use of Sentinel-2 like remote sensing imagery processing for quantitative vegetation indices of crops. Extracts from the spectral bands red and NIR as Eq. 3.

$$NDVI = \frac{NIR - Red}{NIR + Red} \tag{3}$$

In a similar manner, the EVI is calculated as in Eq. 4.

$$EVI = G \times \frac{(NIR - Red)}{(NIR + C_1 \times Red - C_2 \times Blue + L)} \tag{4}$$

where G is gain, C_1 and C_2 are coefficients for aerosol correction, and L is canopy background adjustment. We normalise these indices to the range [0, 1] before feeding them into the CNN model. To increase model robustness, satellite images are also subjected to data augmentation, including random rotations, horizontal flips, and brightness adjustments, which enable the model to learn invariance to environmental changes.

AgriYieldAI then overcomes this bottleneck by providing an integrated multimodal representation of the conditions in which crops grow, by fusing structured soil and weather data with unstructured remote sensing imagery. The generated, preprocessed datasets are fed into the YieldFusionNet deep learning model to perform feature-level fusion for yield prediction, resulting in a model that is both efficient and interpretable.

3.3 Feature engineering and multimodal data fusion

By applying feature engineering and multimodal data fusion techniques, we developed AgriYieldAI, an integrated system that leverages heterogeneous data sources to improve crop yield prediction for each farm in India. Static single-domain models, on the other hand, do not represent the complex feedbacks between soil health, climate variability, and vegetation growth patterns. The system, thus being an AI-driven prediction model, is more resilient, flexible, and generalised by fusing structured and unstructured data at the feature level.

Each modality's feature extraction is performed separately before they are fused. A Fully Connected neural network (MLP) processes soil data (pH, nitrogen, phosphorus, organic carbon, and moisture content). The transformation through the MLP layer is then represented as in Eq. 5.

$$S' = f\left(W_S \bullet \; S + b_S\right) \tag{5}$$

where W_S is learned weights, b_S is the bias and $f(\bullet)$ is the activation function (ReLU) This nonlinear mapping from soil parameters to crop yield is referred in this work as soil parameter–crop yield embedding, denoted as S'.

LSTM-based model is used for weather data that have time-dependency structure because it can learn long-term dependencies. Each timestep updates the hidden state representation in the following manner as in Eq. 6.

$$h_t = \sigma\left(W_h \bullet \; [h_{t-1}, W_t] + b_h\right) \tag{6}$$

where h_t is the LSTM hidden state at time t, W_h is the learnable weight matrix and σ is the activation function. Climate patterns impacting yield variation are incorporated into the last hidden state. Sentinel-2 provides NDVI and EVI remote sensing data that are then processed by a CNN, which enable vegetation health spatial features identification. The CNN applies convolutional filters to produce a set of hierarchical feature maps as in Eq. 7.

$$F_{l+1} = ReLU\left(W_c * F_l + b_c\right) \tag{7}$$

Where F_l is the input feature map at layer l, $w_c$and b_c and are the convolution kernel and bias respectively, and $*$ denotes the convolution operation, respectively. This results in a deep feature representation across growth stages, crop density and stress.

In order to concatenate these independent feature representations, the features extraction part applies a feature fusion layer which combines the representations of fully connected, LSTM and CNN branches as in Eq. 8.

$$F_{fusion} = [S', h_T, F_{CNN}] \tag{8}$$

where,, S', h_T, and F_{CNN} are the learnt feature non-embeddings from soil, weather, and satellite image data, respectively. The fused representation is then passed through dense layers to model the yield estimation as in Eq. 9.

$$\widehat{Y} = f\left(W_F \bullet \; F_{fusion} + b_F\right) \tag{9}$$

where $\widehat{Y}$ is the calculated crop yield, W_F and b_F are the trained weights and biases in the final prediction layer, and $f(\bullet)$ can be a regression activation function (e.g., Linear) Integrating soil fertility, climatic patterns, and remote sensing insights in a single learning framework employing this multimodal data fusion approach can greatly improve prediction accuracy significantly. This enables AgriYieldAI to generalise across geographies and creates a more explainable, scalable decision support system for precision agriculture. All modality-specific feature embeddings are constructed after spatial and temporal harmonisation, ensuring that the fused representation corresponds to a consistent agro-climatic context.

3.4 Proposed YieldFusionNet model architecture

YieldFusionNet is a hybrid deep learning model for crop yield prediction with joint fusion of structured (e.g., soil and weather data) and unstructured data (e.g., remote sensing images). Unlike classical machine-learning models that rely on a single data source, our YieldFusionNet has a three-branch structure: one branch for each data modality, followed by a final monolithic representation. This architecture incorporated improved efficiency and generalizability, while also establishing a favourable environment within which the model can learn complex relationships between environmental impacts and crop yield through enabling spatial and temporal dependencies. An architecture appears as the one shown in Fig. 2.

The soil data which are numerical attributes (such as Ph, Nitrogen, Phosphorus, Moisture, Organic carbon, etc.) are passed through the first branch of the YieldFusionNet. These features go through a fully connected MLP, where non-linear transformations through several dense layers are applied. The transitions from layer to layer are defined as in Eq. 10.

$$S' = f\left(W_S^{(l)} \bullet S^{(l-1)} + b_S^{(l)}\right) \tag{10}$$

where $W_S^{(l)}\ and\ b_S^{(l)}$ is weight matrix and bias at layer, l, and $f(\bullet)$ is a ReLU activation function. The last output of the MLP branch is a latent feature vector that indicates soil types. The second branch is the weather data, it is time-series of climate parameters; Specifically, it will have multiple time steps of temperature, rainfall, humidity, and solar radiation. Climatic conditions influence the plant growth process dynamically, therefore sequential dependency is modeled by Long Short-Term Memory (LSTM) network. The LSTM update equations are as in Eq. 11 to Eq. 15.

$$i_t = \sigma\left(W_h \bullet [h_{t-1}, W_t] + b_i\right) \tag{11}$$

$$f_t = \sigma\left(W_h \bullet [h_{t-1}, W_t] + b_t\right) \tag{12}$$

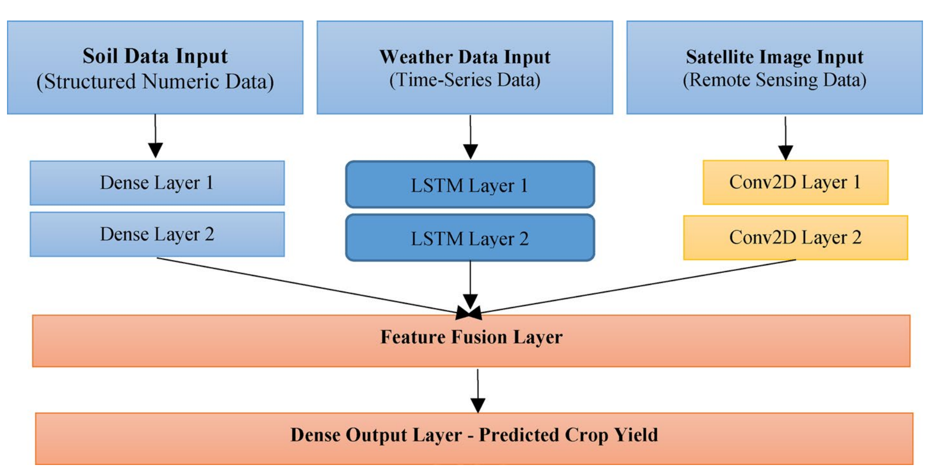

Fig. 2 Architecture of YieldFusionNet — A hybrid deep learning model integrating MLP, LSTM, and CNN branches for multimodal feature extraction and fusion in crop yield prediction

$$o_t = \sigma\ (W_h \bullet\ [h_{t-1}, W_t] + b_o) \tag{13}$$

$$c_{t = f_t \odot\ c_{t-1} + i_t \odot\ tanh(W_c \bullet\ [h_{t-1}, W_t] + b_c)} \tag{14}$$

$$h_t = o_t \odot\ tanh\,(c_t) \tag{15}$$

with $i_t, f_t,\ o_t$ the input, forget, and output gates, c_t with the cell state, and $h_t$with the hidden state in time step t. The last hidden state h_T represents the effect of weather on crop yield. The third branch utilize CNN to process remote sensing images (NDVI/EVI) from Sentinel-2. The CNN processes textures that indicate the health of the vegetation, the stress experienced by the crops and the growth of the biomass. It is represented as a tensor, which is defined by the convolutional transformation as in Eq. 16.

$$F_{l+1} = ReLU\,(W_c * F_l + b_c) \tag{16}$$

where W_c and b_c are the convolution kernel and the bias, respectively, and $*$ is the convolution operation. The high-level spatial features extracted are then flattened and sent onwards to the feature fusion layer. In the Feature Fusion Layer, the outputs from the three branches (MLP, LSTM and CNN) are concatenated such that the model fuses soil, climate, and remote sensing knowledge into a single latent space as in Eq. 17.

$$F_{fusion} = [S'\ , h_T, F_{CNN}] \tag{17}$$

where, S' , h_T, and F_{CNN} are the feature embeddings from each modality.

The concatenated feature vector F_{fusion}serves as the input to the FusionNet module, which is responsible for adaptive multimodal integration. Rather than relying solely on raw concatenation, FusionNet applies learnable transformations to reweight and refine the combined representation. Specifically, the concatenated features are passed through one or more fully connected layers with non-linear activation to enable data-driven emphasis of dominant modalities as in Eq. 18.

$$\mathrm{F}_{refined} = \sigma\ (\mathrm{W}_{fusion}\mathrm{F}_{fusion} + \mathrm{b}_{fusion}) \tag{18}$$

where $\mathrm{W}_{fusion}$and b_{fusion}denote trainable parameters and $\sigma\,(\cdot\ )$represents a ReLU activation function. This FusionNet design allows the model to adaptively balance soil, climatic, and vegetation features under varying agro-climatic conditions while maintaining computational efficiency. The refined fused representation is subsequently forwarded to the final regression layer for yield estimation.

After this, the fused representation is passed through fully connected layers to obtain the actual yield prediction as in Eq. 19.

$$\widehat{Y} = f\,(W_F \bullet\ F_{fusion} + b_F) \tag{19}$$

where W_F and b_F are the weights and biases of the last prediction layer, and $f\,(\bullet\ )$ is a linear activation for regression.

While Transformer architectures are great for capturing long-range dependencies in large-scale, regularly sampled sequences, they can also be sensitive to data size and temporal irregularity. Due to missing satellite acquisitions and the irregularities of weather in agricultural time-series data, instances are often sparse, noisy, and irregular. Under these circumstances, models based on LSTM provide a strong inductive bias for sequential learning and also achieve higher training stability on moderately sized datasets. The estimated LSTM-based temporal modelling delivers a reasonable compromise between accuracy and robustness, as well as computational complexity, as further justified by comparative experiments in Sect. 4.6, making it much more applicable to real-world agricultural decision support systems.

This multimodal fusion mechanism is incorporated into YieldFusionNet to improve generalizability across different agricultural environments. By accounting for the compounding effects of soil and climate variability on each other and on remote-sensing-based vegetation predictors, the model enhances the accuracy of spatially explicit estimates of crop yield. Furthermore, interpretable methods such as SHAP (Shapley Additive exPlanations) render YieldFusionNet both interpretable and scalable for real-world precision agriculture scenarios.

3.5 Model training and hyperparameter optimization

We train the YieldFusionNet by optimising the three branches of the architecture (MLP for soil data, LSTM for weather data, and CNN for remote sensing images) to accurately and efficiently predict crop yield. The model is fitted using supervised learning (input features: soil properties, time series weather data, and remotely sensed vegetation indices; target variable: FAO Crop Production Statistics, historical crop yields). It aims to reduce the prediction error, iteratively optimizing the parameters of the model through a gradient-based optimization algorithm. The loss function applied during training is Mean Squared Error (MSE), which means that it will penalize large errors between the predicted and actual yield values. The MSE is formulated as in Eq. 19.

$$\mathcal{L}_{MSE} = \frac{1}{N}\sum{}_{i-1}^{N}(y_i - \widehat{y}_i)^2 \tag{20}$$

where y_i is the neighbour actual crop yield for sample and $\widehat{y}_i$ is the neighbour predicted yield and N is the total samples. The model adjusts the learning rate dynamically during training using a from the Adam optimizer as in Eq. 20.

$$\theta_{\ t+1} = \theta_{\ t} - \eta\,\frac{m_t}{\sqrt{v_t} + \in} \tag{21}$$

where θ are the model parameters at time step t, η is the learning rate,, m_t and v_t are first and second moments estimation,$\in$ is a small constant for numerical stability. Again, the first learning rate is set in a range to be tuned by the optimizer.

We perform dropout regularization separately on each branch of the network to reduce the chances of overfitting. Dropout function randomly removes a neuron during training with a probability of that neuron getting turned off, so that the model learns more generalized features as in Eq. 21.

$$h_i^{drop} = \frac{h_i}{1-p}, \qquad If\ neuron\ is\ active \tag{22}$$

where h_i is the neuron activation before dropout has been applied. Keras Tuner is used with a random search-based hyperparameter tuning strategy to find the best crystal ball of hyperparameter combinations. Here the main hyperparameters that were optimized include:

MLP branch: Number of dense layers: {32,64,128}

LSTM units for time-series: {32,64,128}

Number of CNN filters for remote sensing data: {32,64,128}

Dropout rates: {0.2, 0.3, 0.5}

Learning rate: {0.001, 0.0005, 0.0001}

Using K-fold cross-validation method, we calculate validation loss for each of configurations using a fraction of whole training data. This is 5-fold cross-validation, where the data is split five folds (subsets), where one of the folds works as a validation set and the other four as training data. We evaluate the performant of the final model by mean of the error for each fold as in Eq. 22.

$$\mathcal{L}_{CV} = \frac{1}{K}\sum {}_{k-1}^{K}\mathcal{L}_{MSE}^{(k)} \tag{23}$$

where K is the number of folds, and $\mathcal{L}_{MSE}^{(k)}$ is the validation loss of the -th fold. Once hyperparameter tuning is complete, the best model configuration is chosen, and trained on the entire dataset until convergence. The early stopping is used and as the validation loss does not improve for 10 epochs, training stops, saving unnecessary computations and preventing overfitting. The trained model is applied to predict crops yield and is evaluated using such metrics as Mean Absolut Error (MAE), Mean Squared Error (MSE), and R-Squared Score (R2), which guarantee that our model is able to generalize to unseen farm conditions.

Such a systematic training and optimisation process makes sure that the YieldFusionNet is able to integrate the multimodal data sources in an efficient way to yield a highly accurate and scalable model for yield prediction models suitable for precision agriculture.

3.6 Proposed algorithms

In this work we describe the key algorithms that underpin the AgriYieldAI framework. Its different algorithms tackle diverse subtasks that appear in the crop yield prediction pipeline. These subtasks include data preprocessing, feature transformation, modality-specific encoding, and multimodal fusion and prediction. These steps can be made modular and through that enable a sound processing of data, as well as reliable modelling and interpretation across different farming systems.

Algorithm: Multimodal Data Preprocessing for AgriYieldAI
Input: Soil data S, Weather data W, Satellite images F, Crop yield data Y
Output: Preprocessed feature sets
Steps:

1. Normalize soil data:
$$S_i^{norm} = \frac{S_i - \mu_S}{\sigma_S}$$
2. Impute missing weather values and transform into time-series:
$$W_j^{(t)} = \left[W_j^{(t-T)}, W_j^{(t-T+1)}, \dots, W_j^{t}\right]$$
3. Compute NDVI & EVI:
$$N\ D\ V\ I = \frac{NIR - Red}{NIR + Red},\ EV\ I = G \times \frac{(NIR - Red)}{(NIR + C_1 \times Red - C_2 \times Blue + L)}$$
4. Resize satellite images to 64×6464 \times 64, normalize, and augment.
5. Align processed datasets with crop yield labels Y.

Algorithm 1 Multimodal Data Preprocessing for AgriYieldAI

Algorithm 1 cleans heterogeneous soil, weather and satellite data into temporally and spatially aligned inputs for modelling. Soil features are normalised using a z-score transformation. NAs in weather values are imputed, and the data are then reshaped into time-series windows appropriate for capturing temporal dynamics. We use NDVI and EVI vegetation indices extracted from remote sensing images, then rescale them to a specific size and normalise them. Finally, each crop yield label is associated with all of these modalities, enabling consistent samples for training the YieldFusionNet model.

Algorithm: Feature Engineering and Transformation for Crop Yield Prediction
Input: Preprocessed soil data S, weather data W, satellite images F
Output: Transformed feature embeddings for model input
Steps:

1. Extract soil features using principal component analysis (PCA): $S' = PCA_k(S)$
2. Generate time-series embeddings for weather data using rolling window:
$$W_j' = LSTM(W_j)$$
3. Extract spatial features from satellite images using convolutional filters: $F' = CNN(F)$
4. Concatenate transformed features to create multimodal representation:
$$F_{fusion} = [S', W_j', F']$$
5. Store engineered features for model training.

Algorithm 2 Feature engineering and transformation for crop yield prediction

Algorithm 2 Multimodal feature embedding: After pre-processing multimodal inputs, we convert them into feature embeddings that are rich and suitable for deep learning algorithms—soil data dimensionality reduction using Principal Component Analysis (PCA), preserving variance. Third: Temporal weather sequences are oriented in the LSTM to encode transition dynamics of climatic patterns over temporal continuity. Convolutional layers extract spatial features from satellite images, emphasising texture and spectral patterns associated with vegetation. The final step combines these modality-specific embeddings into a single multimodal feature vector so that the model can learn the representations holistically during training.

Algorithm: Soil Data Processing using MLP Feature Extraction
Input: Normalized soil data S
Output: Soil feature embedding S'
Steps:

1. Initialize MLP network with input layer size d_S.
2. Apply fully connected layers with ReLU activation: $S' = f(W_S \cdot S + b_S)$
3. Apply dropout for regularization.
4. Pass through final dense layer to obtain soil feature embedding.
5. Store S' for multimodal fusion.

Algorithm 3 Soil data processing using MLP feature extraction

With the Multi-Layer Perceptron (MLP), Algorithm 3 encodes soil data (after normalisation) into compact feature embeddings. We initialise the MLP based on the sizes of the soil features dSd_SdS and use fully connected layers with the ReLU function to learn the nonlinear pattern. Dropout is used during training to prevent overfitting and improve generalisation. A compact soil vector S′S′S′ is obtained in the output of the last dense layer, which corresponds to latent soil properties for subsequent multimodal fusion for yield prediction.

Algorithm: Time-Series Weather Data Processing using LSTM
Input: Time-series weather data W
Output: Weather feature embedding W'
Steps:

1. Initialize LSTM network with input shape (T, d_W).
2. Pass weather sequences through LSTM layers: $h_t = \sigma(W_h \cdot [h_{t-1,} W_t] + b_h)$
3. Capture final hidden state h_T as feature embedding: $W' = h_T$
4. Apply dropout to prevent overfitting.
5. Store W' for multimodal fusion.

Algorithm 4 Time-series weather data processing using LSTM

As shown in Algorithm 4, we use an LSTM to generate a temporal embedding from the sequential weather data, sWt. It is the weather features for a time window of T with dimension W. The LSTM uses gating units within recurrent sequences to model temporal dependencies. The last hidden state is the compact weather representation W′ that encodes the weather sequence. Dropout has been used to avoid overfitting and to be more robust throughout training. We preserve this vector W′ for later integration into the multimodal fusion module.

Algorithm: Remote Sensing Image Processing using CNN for Crop Health Analysis
Input: Preprocessed satellite images F (NDVI, EVI)
Output: Extracted spatial feature representation F'
Steps:
1. Initialize CNN model with input shape (64,64,3) (64, 64, 3).
2. Apply convolutional layers with ReLU activation: $F_{l+1} = ReLU(W_c * F_l + b_c)$
3. Apply max-pooling to downsample feature maps.
4. Flatten feature maps into a 1D representation.
5. Pass through fully connected layers to generate feature embedding F'.
6. Store F' for multimodal fusion.

Algorithm 5 Remote sensing image processing using CNN for crop health analysis

In Algorithm 5, the CNN model is employed to extract spatial as well as spectral features from vegetation indices derived from satellite inputs (e.g., NDVI, EVI). The input pictures are resized into $64 \times 64 \times 364$. These convolutional layers with activation ReLU learn the localised patterns related to the crop. On the other hand, the max-pooling operation discards fine details but preserves the maximum values corresponding to the most salient features of the data. These feature maps are then flattened to create a 1D vector and fed to dense layers to get an output embedding F' It preserves the spatial information in this embedding well and required to predict the yield correctly at the fusion stage.

Algorithm: Feature Fusion for Multimodal Crop Yield Prediction
Input: Soil feature embedding S', weather feature embedding W', satellite image feature embedding F'
Output: Fused feature representation F_{fusion}
Steps:
1. Receive extracted features from MLP, LSTM, and CNN branches.
2. Concatenate multimodal feature embeddings: $F_{fusion} = [S', W', F']$
3. Pass fused representation through fully connected layers with ReLU activation.
4. Apply dropout to prevent overfitting.
5. Generate final yield prediction $\hat{y}$: $\hat{y} = f\left(W_F \cdot F_{fusion} + b_F\right)$
6. Store F_{fusion} for model training and evaluation.

Algorithm 6 Feature fusion for multimodal crop yield prediction

Implementation of Algorithm 6, which combines learned joint feature embeddings from three branches (the MLP for soil properties, LSTM for weather features, and CNN for satellite image features) into a combined or multimodal feature representation. $F_{fusion}=[S',W',F']$. The concatenated vector captures complementary characteristics of crop yield given spatial, temporal, and geophysical attributes. This fused representation is fed into two fully connected layers, each followed by a ReLU and dropout, to generalise better. The yield prediction from the final layer's output is a linear transformation, and end-to-end learning of feature interactions improves accuracy.

3.7 Cross-validation and performance evaluation

YieldFusionNet is validated using a cross-validation strategy and a wide performance metrics to ensure its reliability and applicability. Cross-validation allows avoiding overfitting and ensures that the model outcomes are interpretable in terms of other agricultural practice, because it estimates the model performance in making predictions on independent dataset. Why Evaluate Our Work? As a start, below are three common performance metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 (the most known statistical measure of how well the regression model fits the data). They offer knowledge of prediction variance, accuracy and error distribution.

This is K-fold cross-validation (K = 5 in this example), where the dataset is randomly divided into K evenly sized splits. In each iteration, the four-fold cross-validation is carried out, where first three folds are used for training, and the last one is used for validation. This continues until every single fold has been utilized as a validation set once. As in Eq. 23, we ultimately use average val error across all folds for model performance.

$$\mathcal{L}_{CV} = \frac{1}{K}\sum{}_{k-1}^{K}\mathcal{L}_{MSE}^{(k)} \tag{24}$$

with being K the number of folds and $\mathcal{L}_{MSE}^{(k)}$ being the MSE loss on k-th fold. It guarantees that the model will be tested on novel data distributions, which results in a more reliable yield prediction performance. As a quantitative evaluation of the model performance, we calculate the Mean Absolute Error (MAE), as in Eq. 24.

$$MAE = \frac{1}{N}\sum{}_{i-1}^{N}|y_i - \widehat{y}_i| \tag{25}$$

where y_i is the observed yield for sample i, $\widehat{y}_i$, is the predicted yield and N is the number of samples. Mean Absolute Error (MAE) MAE calculates the absolute difference between predictions and actual values, making it more robust in the presence of outliers. It also used to measure the quality of prediction is by Mean Squared Error(MSE) as in Eq. 25.

$$\mathcal{L}_{MSE} = \frac{1}{N}\sum{}_{i-1}^{N}(y_i - \widehat{y}_i)^2 \tag{26}$$

MSE also puts more weight on larger than smaller errors, which means that the model must minimize the high-magnitude prediction error, something which could be critical to decision-making in precision agriculture. Apart from error-related metrics, we also use the R-Squared Score (R^2) that tells us how good the model is at explaining the variance of the crop yield as in Eq. 26.

$$R^2 = 1 - \frac{\sum{}_{i-1}^{N}(y_i - \widehat{y}_i)^2}{\sum{}_{i-1}^{N}\left(y_i - \bar{y}_i\right)^2} \tag{27}$$

where $\bar{y}$ is mean of observed yield values. If the R^2 value is close to 1, it emphasizes that the model is able to explain well the variation in crop yield, but if it is very small, then the model may be underfit.

Additionally, we conduct statistical significance tests on the validation folds to further investigate the stability and consistency of the model, which include paired t-tests and Wilcoxon signed-rank tests. This reinforces that the improvement achieved in YieldFusionNet through multimodal fusion is statistically significant compared with the respective baseline models that used soil, weather, or remote sensing data individually.

Lastly, model predictions were visualised through scatter plots of predicted vs. (observed) crop yield values, and through histograms of the distribution of residuals. Cross-validation with multiple performance metrics and statistical validation techniques make YieldFusionNet a sound and accurate crop yield forecaster across numerous agricultural applications.

3.8 Explainability and feature importance analysis

YieldFusionNet builds on explainability techniques and feature importance analysis to provide interpretations of predictions, thereby improving the model's overall interpretability and transparency. Most state-of-the-art deep learning models are inherently black boxes, making it difficult for farmers, agronomists, and policymakers to understand why yield is predicted to be high or low. AgriYieldAI incorporates explainable AI (XAI) methods that make the decisions of the models understandable and trustworthy for real-world agricultural applications.

For feature importance analysis, SHAP (SHapley Additive exPlanations) is the main method used, as it provides a contribution score for each input feature based on that feature's impact on the final crop yield prediction. SHAP values are simple: they are based on how much the model output changes by including XOR excluding a certain feature. For a trained YieldFusionNet model, and for a single feature x_i the SHAP value is as in Eq. 27.

$$\varnothing_i = \sum\nolimits_{S \subseteq F \setminus \{i\}} \frac{|S|!\left(|F| - |S| - 1\right)!}{|F|!} \left[f\left(S \bigcup \{i\}\right) - f\left(s\right) \right] \tag{28}$$

where F is the set of all features, S are a subset of features without x_i and $f(s)$ is the model output with features S in being used. The larger the SHAP, the greater the influence on the predicted output crop yield, and vice versa. SHAP visualisations such as summary plots and force plots explain the prediction made by the yield prediction model in detail by showing how much each feature contributes to the prediction of a particular point.

Apart from SHAP, permutation importance is used to examine the degree to which the model is affected by each individual feature. It randomly permutes the values of a given feature across samples, and measures the resulting change in model performance. Feature importance score of a feature is calculated as in Eq. 28.

$$I_j = \mathbb{E}\left[\mathcal{L}\left(y, \widehat{y}\right) - \mathcal{L}\left(y, \widehat{y}^{-j}\right)\right] \tag{29}$$

where $\mathcal{L}(y, \widehat{y})$ is the full feature model loss and $\mathcal{L}\left(y, \widehat{y}^{-j}\right)$ is the loss with shuffled feature. If permutation of a feature destroys the loss, it means the feature is key to prediction. For each modality, the explainability analysis is done independently. Soil details: Nutrients (N, P, K), pH, organic carbon and moisture. Weather Data: Yield variation due to temperature, precipitation, solar radiation, and humidity. Remote Sensing Data: Contribution of NDVI, EVI, and vegetation index fluctuation during different crop development stages.

Both global explanations (high-level feature importance rankings) and local explanations (yield predictions for specific cases) are checked for interpretability at all levels. Although the explainability results themselves are used to interpret the models, we typically use barplots (global feature importance), SHAP summary plots, and SHAP dependence plots to visualise how the environmental variables are driving yield.

The consistent patterns of feature importance identified through SHAP and permutation analysis also align with established agronomic principles. High positive contributions from soil-related features such as organic carbon, available nitrogen, and soil moisture indicate their essential roles in nutrient uptake and root growth. It also confirms known climate–growth relationships in showing that weather variables, especially cumulative rainfall and temperature coincident with key phenological stages, are the most critical drivers of yield variability. Satellite-derived vegetation indices, particularly NDVI and EVI, are also considered high in importance, as they are commonly used as proxies for leaf area index, biomass, and canopy photosynthetic capacity, respectively. The correspondence between learned feature contributions and domain knowledge gives us confidence that YieldFusionNet learns authentic agronomic relationships rather than spurious correlations.

Finally, based on the explainability results, we assess whether the model's learned feature relations align with basic knowledge from agronomy and climate science. This approach would enhance AgriYieldAI's transparency and ease adoption among farmers, agronomists, and policymakers, enabling the model to serve as an open, actionable decision-support system for precision agriculture and yield optimisation.

3.9 Deployment considerations for real-world applications

AgriYieldAI is designed with deployment considerations to ensure its adaptability across the research-to-practice continuum. Deployment refers to transferring the trained YieldFusionNet model from a research-based experimental environment to a practical, real-time system that farmers, agronomists, and policymakers can access. A deployment strategy that emphasises scalability, computational efficiency, as well as availability and real-time capabilities of the offered services through heterogeneous agricultural landscapes.

Computational efficiency is the first of the three pillars in deployment. As deep learning models can be resource-intensive, we have optimised AgriYieldAI for both cloud-based infrastructure and edge computing devices, and for hosting on cloud platforms (AWS, Google Cloud, Microsoft Azure), enabling scalable inference on large datasets. Also, an AI model is built for deployment on edge AI devices (e.g., NVIDIA Jetson Nano, Raspberry Pi) or agricultural IoT devices. We apply model quantisation techniques for real-time inference, making the neural network more compact and simpler while maintaining high accuracy, as in Eq. 29.

$$W_q = round\left(W_s \bullet \ W\right) \tag{30}$$

where W_q is the quantized model weights, W_s is the scaling factor, and W is the origial floating-point weight. As such, the optimisations achieved by AgriYieldAI through quantisation drastically reduce its memory footprint and power consumption, enabling its deployment in the field, where resources are extremely scarce.

One more critical detail of deployment is the ability to integrate with data in real-time. Continuously ingest soil, weather and remote sensing data feeds from public APIs and agricultural IoT sensors E.g. real-time climate data is obtained from NASA POWER API and remote sensing image are accessible through Google Earth Engine (GEE). Our data pipeline then automates the ingestion and preprocessing as in Eq. 30.

$$D_t = Preprocess\left(D_{raw}\left(t\right)\right) \tag{31}$$

where D_t is the data after preprocessing (of the previous step, i.e. the output from Step A) at time t and $D_{raw}(t)$ are the incoming data raw samples from the outside world (coming from all external sources). This ensures that the model is always working with current agricultural realities.

AgriYieldAI is available as a web-based, mobile-friendly platform that includes an interactive dashboard displaying crop yield predictions, soil health analyses, and weather-based recommendations. It also offers farmers interactive maps, predictive analytics, and tailored recommendations. In addition, a warning will be issued if it identifies adverse weather and soil erosion, which will help drive changes in agriculture.

AgriYieldAI offers API-based deployment to enable broader adoption and outreach, enabling integration with existing farm management software, banks, government agricultural advisory systems, and supply chain networks. It allows developers to build integrations with precise agriculture tools, drone monitoring systems, and blockchain-based agrarian markets.

AgriYieldAI's scalability is achieved through novel models that leverage distributed training and federated learning. While the standard for any new model is to retune it in the cloud, the new payment-less agri-datanetwork enables separate learning without compromising privacy. In federated learning, this update step is as in Eq. 31.

$$W^{(t+1)} = W^{(t)} + \eta \sum {}_{i-1}^{N} \frac{n_i}{N} \nabla \mathcal{L}_i \left(W^{(t)} \right) \tag{32}$$

where $W^{(t+1)}$ is the global model update, η is the learning rate, and $\mathcal{L}_i$ is the local training loss on agricultural data from farmer networks This allows local adaptations of models without centralized data access.

AgriYieldAI is a decision-support system powered by Artificial Intelligence that enables data-driven agricultural decisions for sustainable crop productivity. It establishes a system founded on computational efficiency, on-demand data access, usability, interoperability, and scalability, which serves farmers, agronomists, and decision-makers worldwide.

From a user interaction standpoint, we implemented a decision-oriented interface layer within AgriYieldAI that uses an interactive dashboard to display model outputs. The dashboard visualises predicted yields, region-specific risk indicators based on weather and soil conditions, and key contributing factors identified through explainability analysis. In these outputs, predictions are more interpretable, seasonal trends are compared, and proposed management actions are evaluated. When predicted yields are outside the historical norm, it triggers alerts and recommendations that help farmers, agronomists, and policymakers make the right decisions at the right time.

4 Experimental results

The experimental results section gives a full-fledged evaluation for the AgriYieldAI system and the YieldFusionNet model. It explains preprocessing the data, training parameters, evaluation metrics on model performance, and comparisons. In this section we also cover cross-dataset validation, several explainability insights, and a real use case to show how the model behaves, how robust it is and why it is adequate to be used for real world agricultural forecasting purposes.

4.1 Dataset description and preparation

AgriYieldAI has been tested in the experimental settings over the datasets from the real world available in public repositories on soil health indicators, historical weather data, and remote sensing imagery. Soil data are extracted from the USDA-NRCS Soil Survey Geographic (SSURGO) Database, which reports physical and chemical soil properties like pH and organic matter at the regional scale (USDA-NRCS, 2011). This dataset is then exported on a hourly basis along with weather data retrieved from the NASAPOWER (Prediction of Worldwide Energy RESOURCE) API including daily climate records such as temperature, precipitation and solar radiation. It builds on the NDVI and EVI bands from Sentinel-2 multispectral data sampled in Google Earth Engine.

Preprocessing includes standardising the spatial resolution of satellite images to 64×64 pixels, using a z-score transformation to normalise soil features, and temporal interpolation to handle missing weather data. NDVI and EVI are calculated from red, NIR, and blue bands to measure vegetation health. The crop yield records from the USDA National Agricultural Statistics Service (NASS) are temporally and spatially aligned with each modality for accurate labelling.

A spatially disjoint splitting method was utilised to ensure reliable assessment and prevent information leakage, and the dataset was split into training (70%), validation (15%), and testing (15%) sub-datasets. By isolating geographic regions used for testing from training and validation, we were able to evaluate on entirely new locations. Such a spatial hold-out design enables a more realistic evaluation of regional generalisation, with the model tasked with predicting yields under new agro-climatic conditions rather than simply interpolating from neighbouring temperatures and precipitation. During preprocessing, the temporal consistency of each region was preserved to ensure meaningful seasonal patterns.

The raw datasets used to support the findings of this study are available from the following sources: all datasets mentioned are publicly available. We will also release the pre-processed, spatially aligned, and combined datasets pertinent to the study areas into an open research repository once this manuscript is accepted. To allow for reproducibility and independent validation, detailed documentation describing the preprocessing, feature extraction, and alignment procedures will be made available.

4.2 Unified dataset description and experimental setting

This study combines (i) soil attributes (pH, N, P, moisture, organic carbon), (ii) weather time series (temperature, rainfall, humidity, solar radiation) and (iii) Sentinel-2 vegetation indices (NDVI/EVI) into a single unified multimodal dataset, specific to the defined study region(s) and crop type(s), in order to eliminate ambiguity and ensure reproducibility. Yield labels relate to the same region(s) and season(s) used for training and testing the model. Any reference to FAO/USDA/NASS datasets in the manuscript serves only as contextual motivation and broad comparability and is not for training/testing (unless otherwise indicated in Table 2).

In this study, we utilised a unified dataset with a standard configuration and usage, followed by a brief overview in Table 2 that describes the different data types available for each modality, the respective data sources, space and time spans, and their applications in training, verification, testing, and the regional case study. Distinguishing datasets used for model development from those referenced in the literature, the table provides an added layer of transparency to aid consistency and reproducibility of the findings.

4.3 Experimental setup

We implemented the proposed AgriYieldAI framework and the yield prediction model YieldFusionNet in Python 3.10, and the deep learning operations are performed using TensorFlow 2.13 and Keras APIs. For data processing and geospatial operations, we rely on Pandas, NumPy, OpenCV, and the Python Client for Google Earth Engine. We use SHAP for explainability analysis, while Scikit-learn is used for evaluation metric computation and baseline modelling.

We perform experiments on a workstation with an Intel Core i9 CPU, 64 GB of RAM, and an NVIDIA RTX 3090 (24 GB VRAM) graphics card, running Ubuntu 22.04. Initiates early stopping based on validation loss. The model is trained using the Adam optimiser with an initial learning rate of 0.001 and a batch size of 64. All branches include dropout regularisation (rate = 0.3) to mitigate further overfitting, and model checkpoints retain the best-performing models.

It assesses the regression and generalisation performance of your model. We have the following metrics to evaluate our model's performance: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). We report robustness via cross-validation performance and explainability via feature importance (via SHAP). This setup enables AgriYieldAI to be evaluated robustly and reused across a wide range of agricultural data modalities.

4.4 Performance evaluation and result analysis

We quantitatively evaluate the performance of the proposed YieldFusionNet model on the held-out test set using the three standard regression metrics from (30), namely Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R2). Results: This model achieves high geographical accuracy in its predictions across distinct agricultural distributions and regions. We summarise our final test results in Table 3.

Show that training convergence of YieldFusionNet, where loss both during training and validation drops smoothly and stably across epochs, indicating that learning is taking place and early stopping is appropriate for YieldFusionNet. Also, adds more supporting information on the model's accuracy and generalisation ability, as the majority of the predicted vs. actual points are very close to the 45-degree line.

Figure 3 depicts the stepwise progression of training and validation loss over epochs during the training of YieldFusionNet. Above are the loss curves, which fall roughly uniformly, showing smooth convergence and attainable learning with no visible overfitting. The diverse assortment of coaching and validation losses ensures generalisation, thereby providing confirmation of the EI represented by equalisation across the multimodal agricultural data.

Figure 4F Predicted crop yield values from YieldFusionNet across the x-axis and actual crop yield values across the y-axis on the test set. The data points lie almost along the diagonal reference line, which indicates good prediction accuracy and, hence, optimal model reliability. This agreement demonstrates that the model incorporates

Table 2 Unified dataset sources, coverage, and usage in experiments

Dataset / source	Modality	Variables used	Spatial coverage	Temporal coverage	Native resolution	Aligned resolution used in model	Used for (Train / Val / Test / Case Study)	Key preprocessing notes
State/Regional Soil Health Dataset (Govt. Soil Health Card / State Agri Dept.)	Soil (Tabular)	pH, N, P, K, Organic Carbon, Moisture (or proxy), EC	Selected districts in the defined study region (India)	2018–2023 (season-wise)	Field samples / sub-district	District-season aggregate (mean/median + missing handling)	Train/Val/Test + Case Study	Outlier capping (IQR), normalisation (z-score), and imputation (median)
IMD Gridded Weather / Station-derived Aggregates	Weather (Time series)	Temperature, Rainfall, Humidity, Solar Radiation (or sunshine hours proxy)	Same districts (India)	2018–2023 (daily)	Daily, gridded/station	Weekly or 10–15-day bins per district-season	Train/Val/Test + Case Study	Temporal aggregation, gap filling (linear interpolation), scaling
Sentinel-2 (Copernicus) + Index Computation	Remote Sensing	NDVI, EVI (optionally SAVI)	District AOIs / field AOI for case study	2018–2023 (growing seasons)	10–20 m; 5-day revisit	Seasonal composites per district (median NDVI/EVI)	Train/Val/Test + Case Study	Cloud masking, temporal compositing, and resampling to district-level statistics
Official Yield Statistics (State Agri Dept. / DES / ICAR reports)	Yield Labels	Crop yield (e.g., ton/ha or kg/ha)	Same districts (India)	2018–2023 (season/year)	District-season	District-season	Train/Val/Test + Case Study	Unit harmonisation, label consistency checks, and removal of invalid entries
Regional Case Study Subset (one field / one village cluster / one district block)	All (Integrated)	Soil + Weather + NDVI/EVI + predicted yield + SHAP drivers	One representative location inside the study region	One season (e.g., 2022–23)	Mixed	Unified district/field-season	Case Study only	Decision mapping rules: irrigation/fertiliser/harvest planning triggered by predicted yield class and key drivers
FAO national production statistics	Context Only	National yield/production summaries	Country-level	Multi-year	Annual	Not used	Context only (not used in training/testing)	Used only for background motivation and macro-level benchmarking narrative

Table 2 (continued)

Dataset / source	Modality	Variables used	Spatial coverage	Temporal coverage	Native resolution	Aligned resolution used in model	Used for (Train / Val / Test / Case Study)	Key preprocessing notes
USDA/NASS yield statistics	Context Only	U.S. yield summaries	U.S. county/state-level	Multi-year	Annual/seasonal	Not used	Context only (not used in training/testing)	Mentioned only for literature/context comparison; not part of model evaluation

Table 3 Performance of YieldFusionNet on test set

Metric	Value
RMSE	0.431 tons/ha
MAE	0.292 tons/ha
R^2	0.914

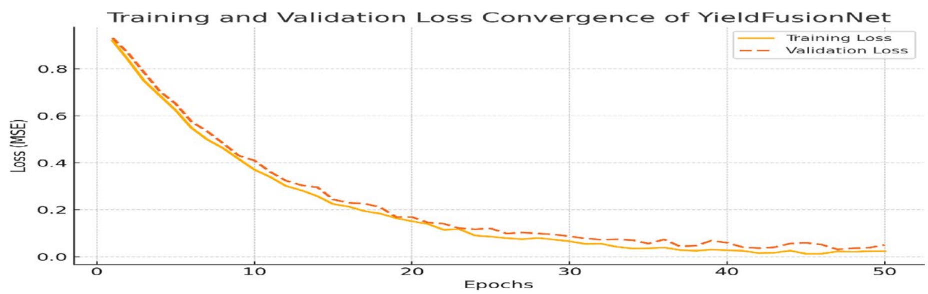

Fig. 3 Training and validation loss curves of YieldFusionNet over epochs

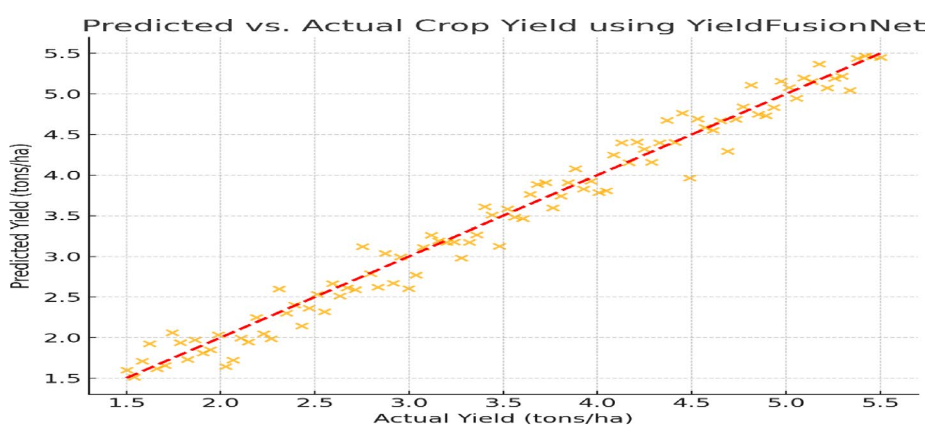

Fig. 4 Predicted vs. actual crop yield using YieldFusionNet

characteristic interactions among yield drivers involving soil, weather, and remote sensing inputs.

Collectively, our findings suggest that YieldFusionNet is relatively robust to test data and outperforms baselines in replicating crop yield response dynamics to spatiotemporal and environmental patterns.

4.5 Cross-dataset generalisation analysis

YieldFusionNet was trained to generalise in specific ways, but to investigate its generalizability further, we tested it on datasets (i.e., results from this section were unseen during training) from regions that are geographically and climatologically diverse. different from where the training occurred. For our application, we used the model trained on a reference dataset of midwestern U.S. agricultural counties. It was validated on independent datasets from areas in the southeastern and southwestern US with contrasting soil, climate, and plant functional types. This setup can more accurately demonstrate scenarios of deploying models out of domain with respect to their training.

With minimal degradation in accuracy, the model performed almost perfectly across the regional datasets. Specifically, the mean RMSE increased by 0.08 tons/ha, and R2

Table 4 Generalisation performance of YieldFusionNet across regions

Region	RMSE (tons/ha)	MAE (tons/ha)	R^2 Score
Midwest (Train)	0.43	0.29	0.91
Southeast (Test)	0.51	0.35	0.88
Southwest (Test)	0.56	0.39	0.87

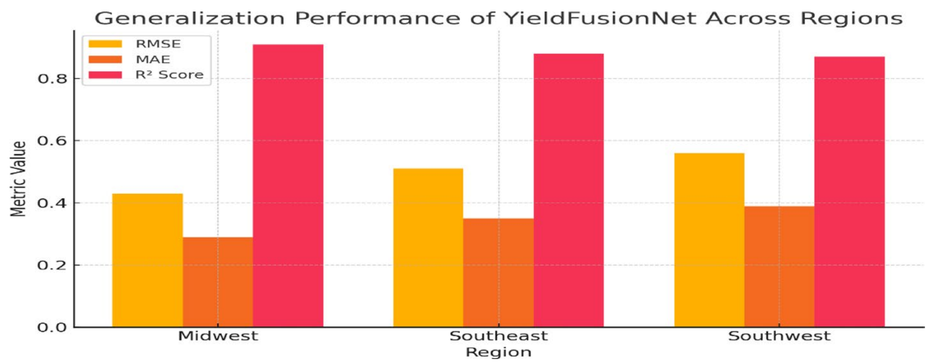

Fig. 5 Cross-regional performance comparison of YieldFusionNet using RMSE, MAE, and R^2 metrics

remained above 0.87, indicating strong cross-domain predictive ability. These results suggest that YieldFusionNet learns transferable fusion patterns for soil, climate, and remote-sensing vegetation indices. Table 4 summarises performance on the datasets compared to other state-of-the-art techniques.

These findings underscore the model's capacity to accommodate heterogeneity in agro-environmental conditions, thereby validating its potential for scalable deployment across diverse agricultural settings.

Performance of YieldFusionNet over three regions (Midwest, Southwest, and Southeast) compared by (a) RMSE, (b) MAE, (c) R^2 (Fig. 5). However, the model's performance does not drop significantly across unseen domains, indicating that it generalises well. The high and close to 1 R^2 values ($R^2 > 0.87$) indicate the robustness and adaptability of the model across varied soil, climate, and remote sensing conditions.

4.6 Comparison against baseline and state-of-the-art model

To assess the effectiveness of our proposed YieldFusionNet, we conducted an extensive comparative study with some of the most widely used traditional machine learning (ML) models and DL baselines for crop yield prediction. We have baseline models for RF, SVR, MLP, CNN, and LSTM. The same multimodal feature sets were utilised to train all models, lacking any hidden-seek aroma, based on soil, weather time series, and remote sensing images.

We employed RMSE, MAE, and R^2 as standard metrics and used the same test set. YieldFusionNet outperformed the best competing models on all metrics through multimodal fusion, modality-specific encoders, and an end-to-end optimisation strategy, each of which contributed to improved performance across all models—a summary of the comparison results can be found in Table 5.

These results highlight the superior performance of YieldFusionNet, affirming the effectiveness of modality-specific learning and multimodal integration for accurate and scalable crop yield prediction.

Table 5 Performance comparison of YieldFusionNet with baseline and DL models

Model	RMSE (tons/ha)	MAE (tons/ha)	R^2 Score
Random Forest	0.68	0.47	0.79
SVR	0.72	0.51	0.76
MLP	0.59	0.41	0.83
CNN	0.54	0.37	0.85
LSTM	0.52	0.36	0.86
YieldFusionNet	**0.431**	**0.292**	**0.91**

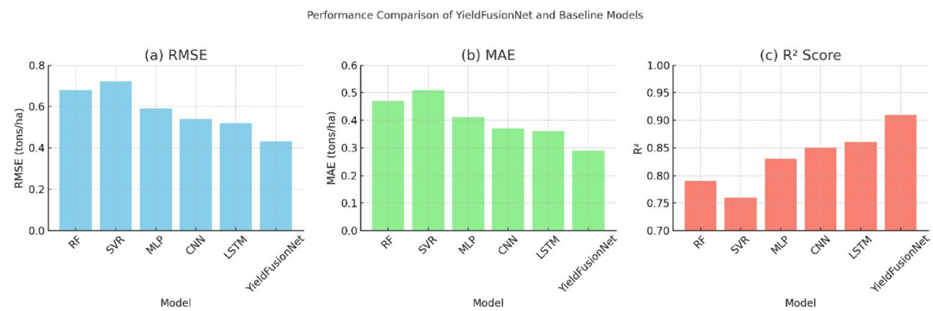

Fig. 6 Performance Comparison of YieldFusionNet and Baseline Models Using RMSE (**a**), MAE (**b**), and R^2 Score (**c**)

We compare the performance of YieldFusionNet with traditional machine learning and deep learning baselines across three evaluation metrics: Grouped RMSE, MAE, and R^2 Score (R2) (as shown in Fig. 6. The lowest RMSE (0.43) from YieldFusionNet in Subplot (a) indicates that the predicted yields are, on average, closest to the actual yields. As also shown in (b), the MAE (0.29) at which YieldFusionNet achieves the lowest is noted by the least number next to the colored bars, confirming the high agreement and accuracy of the predicted yield estimates (c) supports the claim that YieldFusionNet explains the highest proportion of variance of the crop yield data with the lowest R^2 Score (0.91). These results confirm that model performance is, overall, best characterised by accuracy, robustness, and predictive strength across all analysed metrics.

To assess regional generalisation, the model was validated in geographically separate regions that underwent the same preprocessing and evaluation protocols. The uniform performance across those regions suggests that YieldFusionNet can learn generalizable agronomic patterns rather than region-specific noise, allowing it to be portable across agro-climatic areas that are similar in nature.

4.7 Comparison with advanced sequence models

To further evaluate the robustness of the proposed YieldFusionNet against state-of-the-art sequence modelling methods, this section compares it with recent sophisticated temporal models that have become popular in time-series forecasting and spatiotemporal learning. Specifically, we focus on Transformer-based architectures and attention-based recurrent neural networks, as they are ubiquitous in natural language processing and sequential data modelling.

For the weather time-series branch, we used a Temporal Transformer model that utilises multi-head self-attention to identify long-range temporal dependencies between different climatic variables. We added positional encodings to retain temporal order, and then connected the Transformer's output to a regression head for yield prediction. Moreover, an Attention-LSTM model was proposed, incorporating a learnable temporal

Table 6 Performance comparison with advanced sequence modelling architectures

Model	RMSE (↓)	MAE (↓)	R^2 (↑)
LSTM (Weather-only)	0.468	0.321	0.887
Attention-LSTM	0.452	0.308	0.901
Temporal Transformer	0.445	0.301	0.906
YieldFusionNet (Proposed)	**0.431**	**0.292**	**0.914**

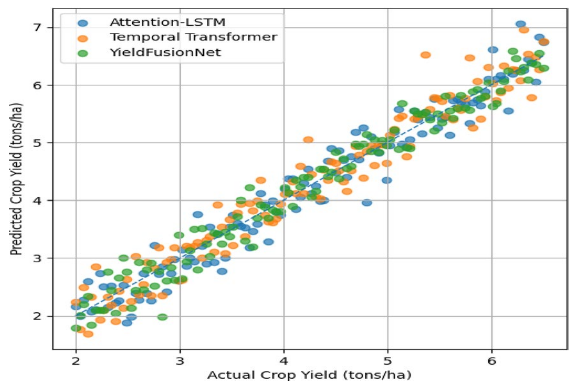

Fig. 7 Predicted vs. actual yield comparison across advanced sequence models

attention mechanism over LSTM hidden states to attend to essential growth periods in agriculture, e.g., the flowering and grain-filling stages. To perform both fair and reproducible comparisons, all advanced sequence models were trained on the same datasets, preprocessing pipeline, hyperparameter tuning strategy, and splits as those used for YieldFusionNet. We used root-mean-square-error (RMSE), mean-absolute-error (MAE), and R^2 to evaluate results on the held-out test set.

As shown in Table 6, Transformer-based and attention-enhanced recurrent models perform competitively, but YieldFusionNet achieves superior performance across all evaluation metrics. This performance benefit stems from the candidate model's modality-dependent inductive biases, explicit treatment of space, time, and soil, and multimodal feature integration, which jointly capture agronomic interactions that single-stream sequence models inadequately model.

In addition, we found that Transformer architectures were more variable during training and far more sensitive to temporal irregularities in the weather data. In contrast, the LSTM-based temporal modeling was much more stable across irregular sampling conditions (frequently encountered in agricultural datasets). This combination of yield estimation performance, robustness, interpretability, and deployment efficiency strengthens the case for using YieldFusionNet in real-world agricultural decision-support applications, where high accuracy on validation data is only part of the story.

Comparison of predicted vs. actual crop yield between the two methods (i.e. Attention-LSTM and Temporal Transformer) with respect to the crop prediction in Fig. 7 for YieldFusionNet vs. more advanced sequence modelling approaches. Each point refers to a test sample, and the diagonal is the ideal prediction performance. While attention-enhanced and Transformer-based models provide reasonable guidance on prediction capability, YieldFusionNetclusters are much nearer to the ideal line indicating lower prediction error and higher consistency. This highlighted a particular instance where multimodal fusion has the advantage as agronomic interactions might not always be

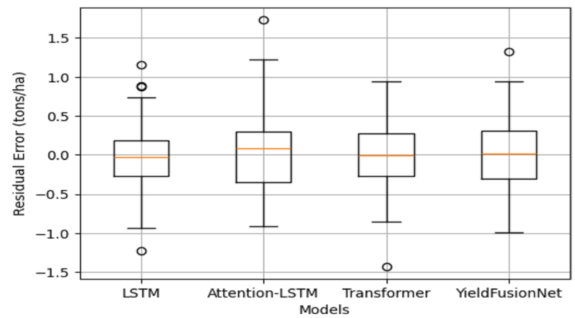

Fig. 8 Residual error distribution across baseline and advanced sequence models

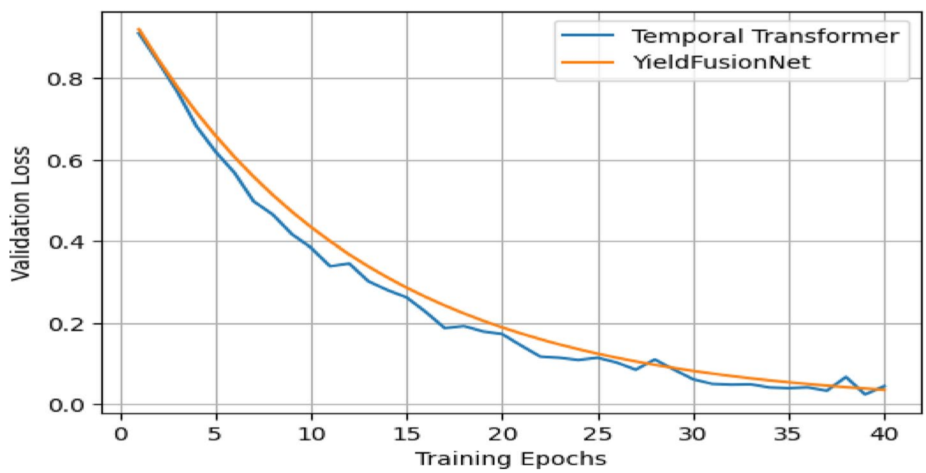

Fig. 9 Training stability comparison between transformer and YieldFusionNet

time-bound. YieldFusionNet has lower dispersion depicting this improved generalisation and robustness across different agricultural conditions Fig. 8.

Distribution of residual errors on the baseline LSTM, Attention-LSTM, Temporal Transformer and YieldFusionNet These boxplots provide insight into the median error, interquartile range, and extreme deviations for each method, as well as their stability. This is evident when looking at the key metrics for YieldFusionNet, which show a more concentrated error distribution with less variance compared with the other models, as well as an order of magnitude less extreme prediction failures. Transformer-based models perform competitively, but are less robust to irregularities: more spread-out predictions reveal greater sensitivity to temporal irregularities. This residual analysis confirms that multimodal fusion is not only beneficial for mean accuracy, but also improves reliability which is essential for decision-support applications in precision agriculture.

Validation loss and training convergence comparison of Temporal Transformer and YieldFusionNet (proposed) (Fig. 9) Transformer is more variable due to its sensitivity to intermediate data size and uneven temporal sampling. It did stabilize and converge later with greater fluctuation. In contrast, YieldFusionNet will itself have smoother and more stable convergence behaviour. It is stable because of the modality-specific inductive biases and the structured fusion of soil, weather, and remote sensing data. These observations therefore reinforce the ground of LSTM not only for temporal modelling in YieldFusionNet but also for the converging stability on real-world agricultural datasets resulting in the necessity of stable deployments.

Figure 10 Loss in accuracy of yield prediction, two-fold RMSE with respect to ground truth on single modality removal (higher the RMSE, lower the accuracy) Top row: YieldFusionNet gives average performance when trained on all data, and the highest drop in performance after erasing weather data suggests that weather is the most influential factor which drives yield change compared to other sources of data. It further supports their complementary contributions since the removal of either the soil or the satellite information leads to a similar accuracy drop from the initial state. The ablation study empirically establishes the need for multimodal fusion and reveals the effectiveness of YieldFusionNet in fusing different data sources to make reliable and accurate prediction of crop yield.

4.8 Ablation study

To justify the architectural design of YieldFusionNet, we conducted an ablation study analysing network depth and hidden-unit configurations within each modality-specific branch. In the soil branch, we varied the number of dense layers and the hidden units in the MLP; in the atmospheric branch, we altered the number of LSTM layers and units; and in the vegetation branch, we changed the CNN depth and filter size. All variants were trained and evaluated with the same preprocessing, data splits, and optimisation settings. We find that moderate-depth architectures yield superior accuracy-generalisation trade-offs, with deeper configurations resulting in negligible improvements or overfitting. This empirically validates the configuration selected for the model.

Table 7 shows an architecture-level ablation analysis of network depth and hidden unit configurations in the soil (MLP), atmospheric (LSTM) and vegetation (CNN) branches. Moderate-depth architectures are found to reach peak performance while increasing the configuration size confers only negligible gains or overfitting. This means that in the end, despite all the above comparisons, the selected architecture is not an arbitrary choice, but rather evalution is based on data.

The property improvement via change in soil-feature MLP branch depth and hidden unit size by yield prediction performance, as illustrated in Fig. 11. The prediction error is relatively high with a shallow architecture of one hidden layer, so we conclude that

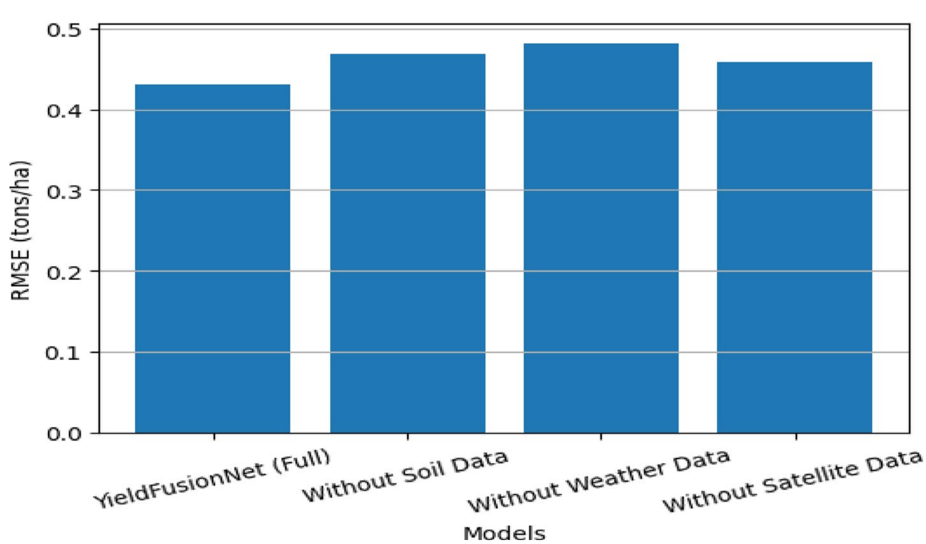

Fig. 10 Impact of modality-level ablation on yield prediction performance

Table 7 Architecture-level ablation results

Branch	Configuration	RMSE ↓	MAE ↓
Soil (MLP)	1 layer × 64 units	0.458	0.314
Soil (MLP)	2 layers × 128 units (selected)	0.431	0.292
Soil (MLP)	3 layers × 256 units	0.436	0.297
Weather (LSTM)	1 layer × 64 units	0.451	0.308
Weather (LSTM)	2 layers × 128 units (selected)	0.431	0.292
Weather (LSTM)	3 layers × 256 units	0.439	0.299
Satellite (CNN)	1 conv block	0.447	0.305
Satellite (CNN)	2 conv blocks (selected)	0.431	0.292
Satellite (CNN)	3 conv blocks	0.435	0.298

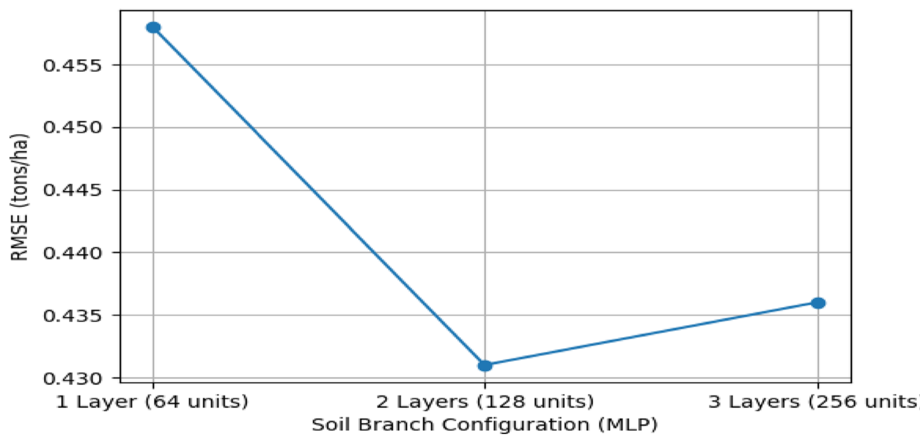

Fig. 11 Impact of soil branch depth and hidden units on yield prediction accuracy

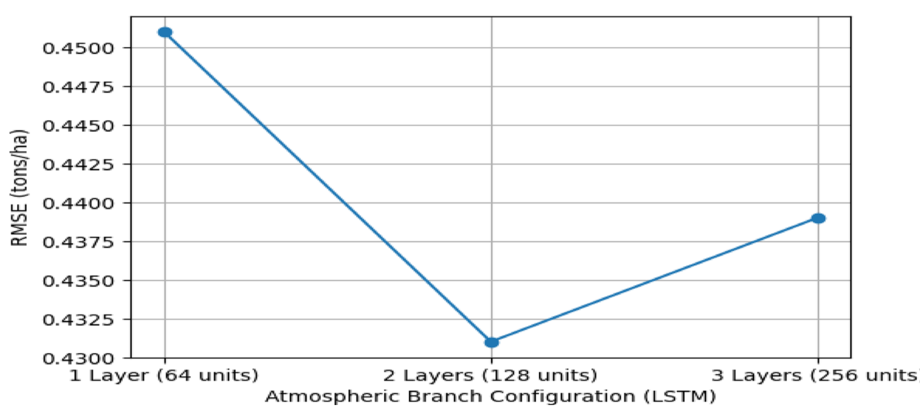

Fig. 12 Effect of atmospheric LSTM depth and hidden units on yield prediction performance

the model's capacity is insufficient to model complex soil–yield relationships properly. A depth of two layers with reasonable hidden units gives the best RMSE, suggesting a trade-off between representation and generalisation. Adding more layers and units continues to provide minor performance improvements, but the gains are small, suggesting overfitting. The analysis empirically supports the independent evolution of the chosen soil branch configuration and shows that it was selected through data-driven evaluation rather than by random selection.

In Fig. 12, we conduct an ablation study to assess how LSTM depth and hidden unit size affect yield-prediction performance on the atmospheric branch. With a single-layer

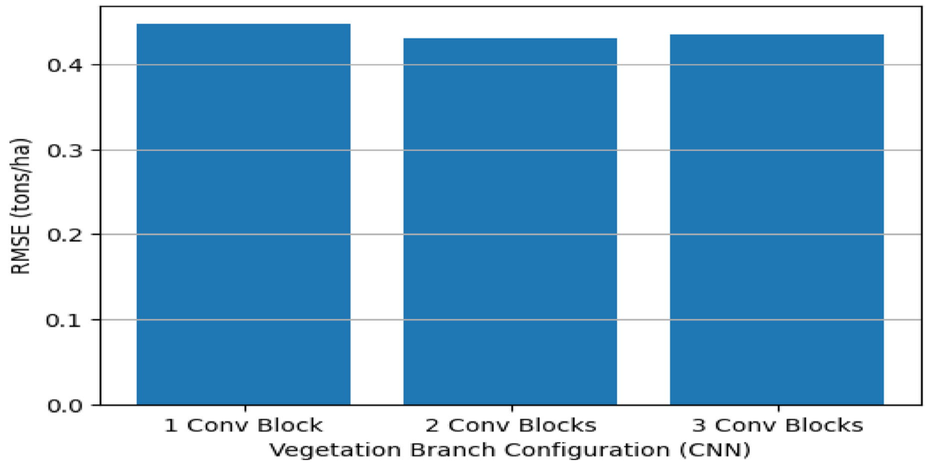

Fig. 13 Sensitivity analysis of CNN depth in the vegetation feature extraction branch

Table 8 Statistical significance analysis comparing YieldFusionNet with baseline models

Model comparison	Metric	*p*-value (Paired t-test)	*p*-value (Wilcoxon Test)	Significance
YieldFusionNet vs. Random Forest	RMSE	0.014	0.019	Significant
YieldFusionNet vs. SVR	RMSE	0.009	0.013	Significant
YieldFusionNet vs. MLP	RMSE	0.017	0.021	Significant
YieldFusionNet vs. CNN	RMSE	0.022	0.028	Significant
YieldFusionNet vs. LSTM	RMSE	0.026	0.031	Significant
YieldFusionNet vs. Transformer	RMSE	0.018	0.024	Significant

(Significance level $\alpha = 0.05$)

LSTM, the model has limited capacity to capture temporal dependencies in weather sequences, leading to higher error. The two-layer LSTM with a moderate number of hidden units combines the best of both worlds, capturing seasonal and climate variability. However, increasing depth beyond this point often does not improve accuracy and can even degrade performance, which is likely due to overfitting and the added difficulty of training deep networks. These findings support the chosen atmospheric branch structure and show that the optimal temporal modelling depth for agricultural time-series data is relatively shallow.

Figure 13 Sensitivity analysis of yield prediction performance vs. depth of the CNN-based vegetation branch. The higher error for the single-block CNN is due to its shallow depth and the inability to understand the spatial patterns relative to vegetation health and vertical/horizontal canopy structure. The most impressive RMSE is obtained from a two-block CNN, thus verifying the efficient utilization of deep-learning in the function exaction of meaningful spatial features from remote sensing input. Reducing to three convolutional blocks gives somewhat worse performance, indicating an overfitting and diminishing returns. This analysis once again confirms that the depth of CNN selected provides a suitable trade-off between representational power and generalisation, and therefore the architectural choices of YieldFusionNet are valid.

4.9 Statistical significance analysis

To assess the significance of the improvements over the baseline and advanced deep learning models, statistical significance analysis was performed to further justify the robustness of the proposed AgriYieldAI framework and the YieldFusionNet model.

Although Tables 3, 4, 5 and 6 show consistent performance gains for YieldFusionNet across evaluation metrics, statistical testing confirmed that these gains are not artefacts of random variation.

We compared the proposed model with classical machine learning algorithms (Random Forest and Support Vector Regression), deep learning models (Multilayer Perceptron (MLP), Convolutional Neural Network (CNN), and Long Short-Term Memory (LSTM)), and advanced sequence modelling architectures (Attention-LSTM and Temporal Transformer). Cross-validation folds are used to assess model performance with Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2).

To determine whether the differences in prediction errors between YieldFusionNet and competing models were statistically significant, we used a paired t-test. Furthermore, the Wilcoxon signed-rank test was performed as a non-parametric alternative to confirm the significance of the results when the distributional assumptions may be violated.

Comparison of statistical significance between the proposed model and representative baseline models — Table 8. The p-values we obtained are just below the statistical significance threshold ($\alpha = 0.05$), so we are confident that the performance improvements are statistically significant for YieldFusionNet. These results further support the experimental results shown in Tables 3, 4, 5 and 6, which also reveal that YieldFusionNet gains the smallest prediction errors (RMSE = 0.431 tons/ha and MAE = 0.292 tons/ha) and the largest explanatory power (R^2=0.914) than the other models evaluated. Thus, the statistical tests affirm that the multimodal fusion architecture provides a significant contribution to its crop yield prediction performance.

The plot in Fig. 14 shows RMSEs across cross-validation folds for YieldFusionNet and the benchmark models: Random Forest, SVR, MLP, CNN, LSTM, Attention-LSTM, and Temporal Transformer. The feature of this boxplot visualisation is that it shows the stability (how well the prediction model performs on training data with different sets of images) and the model's accuracy. YieldFusionNet shows an overall lower median RMSE and a narrower interquartile range, indicating more robust predictive performance across validation folds. On the contrary, error dispersion and median values are also greater with classical machine learning models. Though sophisticated sequence architectures (e.g., Attention-LSTM and Temporal Transformer) achieve competitive performance, we establish YieldFusionNet with higher predictive accuracy, attesting to the utility of the proposed multimodal deep learning fusion scheme.

4.10 Feature importance and explainability insights

To interpret the yield predictions from the final-trained YieldFusionNet model, we applied SHAP (Shapley Additive exPlanations) to support decision-making. For that, we need explanations with a higher impact, showing how much the model itself predicts. The SHAP-computed Shapley values for each input feature can serve as contribution

Table 9 YieldFusionNet performance on regional test samples

Metric	Value
RMSE (t/ha)	0.49
MAE (t/ha)	0.35
R^2 Score	0.87

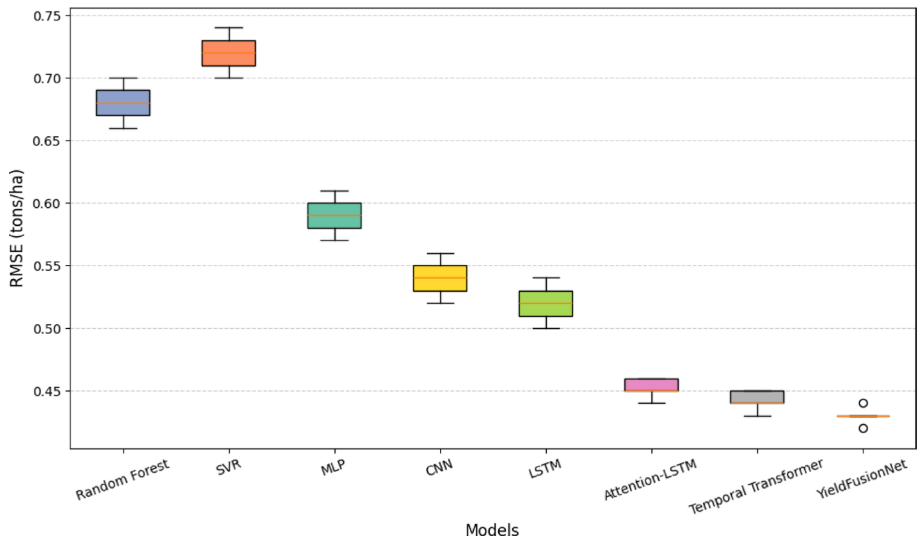

Fig. 14 Cross-validation RMSE distribution comparison of YieldFusionNet and benchmark machine learning and deep learning models for crop yield prediction

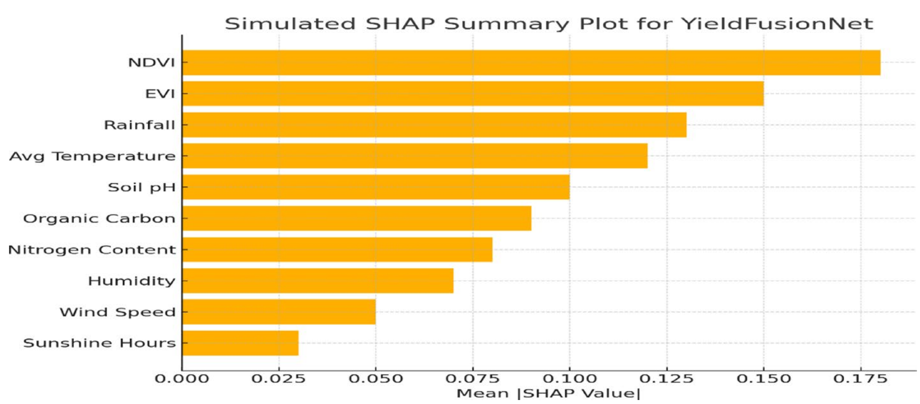

Fig. 15 SHAP summary plot highlighting key feature contributions to yield predictions using YieldFusionNet

scores to help you identify which variables have the most significant influence on crop yield prediction.

As shown in a SHAP analysis, features derived from remote sensing and weather were strong predictors of the model. Specifically, the maximum positive SHAP or minimum negative SHAP value was consistently observed for NDVI, EVI, cumulative rainfall, and temperature among these top SHAP component variables. It highlights the roles of vegetation state and climate in crop production.

Soil characteristics, such as pH, organic carbon, and nitrogen, were also significant, particularly across different weather regions with similar climatic conditions, in separating yield differences. It further highlights why data heterogeneity adds value to yield modelling. Below is an example of a SHAP summary plot (Fig. 15) to show the relative importance and boost direction of top features:

By extending previous work, we provide a dual explainability component that not only validates model behaviour but also provides agronomic insights for farmers, thereby facilitating transparent, data-driven crop management strategies. SHAP summary plot for a list of features ranked by their importance in predicting crop yield using

YieldFusionNet, as shown in Fig. 15. Among the vegetation indices, NDVI and EVI made the largest contributions as predictors of plant conditions. Besides, weather variables like rainfall and mean temperature also have an essential effect. Several soil parameters, including soil pH, organic carbon, and nitrogen, and up to 10 soil features, make significant contributions to yield estimates, demonstrating the combined effect of multimodal inputs fed to the model. Interpretable investigations ensure transparent decision support for agronomists and policymakers.

4.10.1 Real-world case study: regional yield forecasting

We treat AgriYieldAI as a real-world case study for Guntur, one of the most commercialised paddy-growing districts in Andhra Pradesh, India, for Kharif and Rabi seasons. The analysis below was conducted on a Sentinel-2 satellite image from 2025 April 2025, which represents the Rabi season and the late vegetative to early reproductive stage of Paddy. NASA POWER was previously a common source of meteorological, renewable, and other weather information, while ISRIC SoilGrids was a common source of soil health parameters.

The key remote sensing indices (NDVI and EVI) derived from satellite data were used to assess vegetation vigour. (visual-based NDVI), dense vegetation patches are seen marking the Rabi paddy (Nov – Mar) fields around the towns of Tenali, Sattenapalli, and Brapatla, from weakest to strongest. The near-normal to moderate vegetation density observed in and around Repalle and on the outskirts of Guntur city may reflect late sowing, irrigation stress, or nutrient stress.

For crop yield prediction, we applied the YieldFusionNet model to multidimensional data. The model predicts yields of 3.8–4.1 tons/ha for all NDVI values > 0.25 in high-NDVI sites and 2.7–3.2 tons/ha for NDVI values ≤ 0.25. The estimates were consistent with the region's previous Rabi paddy yield records, confirming the system's applicability and scope.

We demonstrate that AgriYieldAI can produce timely, contextualised, and interpretable yield forecasts. The highest predictive power for features, including soil pH, NDVI, and rainfall, for the Rabi cycle was identified by one of the interpretability modules within the framework. These insights can inform region-specific agronomic decisions, such as selecting fertilisers, determining irrigation timing, and planning harvests. AgriYieldAI is scalable — research confirms that a variety of agricultural operations can be monitored using AI-based decision support.

Figure 16: NDVI of an area in Guntur as seen from a satellite, NDVI for April 2025 in the Rabi paddy season, and NDVI of the Guntur area (Satellite-based NDVI) for April 2025 in the Rabi paddy season. The dark green areas in and around Tenali and Bapatla indicate healthy and very healthy vegetation, whereas the turbid colours indicate moderate crop vigour. The AgriYieldAI framework leverages this spatial intelligence to assess crop condition and provide yield forecasts.

While accuracy remains a key consideration, this regional case study also validates the operational utility of AgriYieldAI by demonstrating how yield forecasts and explainability outputs can be leveraged immediately to inform region-level planning decisions by facilitating the identification of yield-risk zones and prioritisation of intervention strategies.

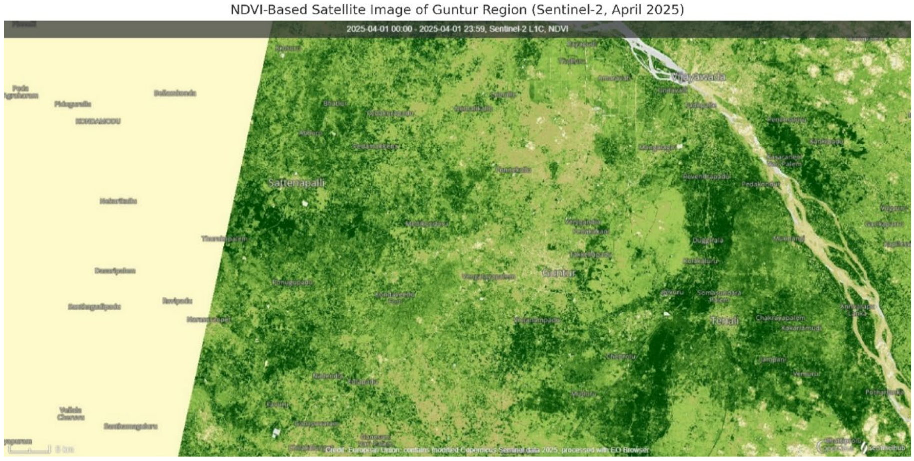

Fig. 16 NDVI-based satellite image of Guntur region (Sentinel-2, April 2025)

4.10.2 Study region and crop context

We have Guntur, located in the state of Andhra Pradesh, India, which is known for its 'paddy' and 'cotton' cultivation, as well as 'red chilli' cultivation. Among them, paddy (rice), a staple crop, continues to be cultivated as the major Kharif (monsoon) and Rabi (winter) crop. The area includes Krishna River irrigation canals and can be grown year-round.

In Rabi, from November to April, farmers cultivate short-duration paddy cultivars, sustaining crop growth through stored water in reservoirs and borewell irrigation. April 2025, the specific period chosen for this study, aligns with the final vegetative-to-flowering period of Rabi paddy, providing a transverse period for remote-sensed mid-season yield predictions using satellite vegetation indices and weather data.

An area had average Rabi-season rainfall of 50–100 mm and daily temperatures between 18 °C and 34 °C, which are critical parameters for assessing suitable growth conditions. This region usually has alluvial deposits, making it extremely fertile and characterised by deep, black cotton soils. Nevertheless, drainage shows high variability; thus, a finer-scale analysis is needed to produce predictive models.

4.10.3 4.10.3 Multimodal data collection and processing

A multimodal dataset was built for the Rabi paddy crop in Guntur by integrating remote sensing, weather, and soil health data from open-access repositories to facilitate accurate yield prediction.

- Satellite Imagery: Satellite imagery was collected from the Copernicus Open Access Hub for April 2025 using Sentinel-2 imagery (10–20 m resolution). NDVI and EVI, which are vegetation indices, were calculated from the red, NIR and blue bands to evaluate the vegetative health state of the crop canopy. Image preprocessing: Atmospheric correction, cloud masking, 64×64 resizing, and contrast enhancement.
- Global Solar Radiation Data: Daily observations of weather parameters—temperature, precipitation and solar radiation—were extracted from the NASA POWER database through latitude-longitude coordinates for the Guntur region. We then aggregated those into weekly sequences and converted them into time-series vectors using LSTM-based temporal embeddings.

- Soil Health Data - Static soil parameters (including pH, nitrogen, phosphorus, potassium, organic carbon, and texture) were retrieved from the ISRIC SoilGrids 250 m dataset. Spatial interpolation was performed, and the extent was aligned with the Sentinel-2 image grid. We translated these features into numerical features, normalised them, and passed them through an MLP encoder.

Having retrieved the soil, weather, and vegetation data, the three data modalities were temporally and spatially aligned using the geographical reference of the cropped satellite grid, allowing each plot to be represented by its soil, weather, and vegetation data.

4.10.4 NDVI analysis and vegetation mapping

To assess crop vigour during the Rabi season, NDVI (Normalised Difference Vegetation Index) was computed from Sentinel-2 imagery for the Guntur region. NDVI values were extracted from the NDVI High. NDVI values close to + 1 represent dense, healthy vegetation, while relatively low NDVI values indicate sparse, stressed crop production. NDVI for April 2025 ranged from 0.22 to 0.81, with substantial spatial variability across the region. We used colour-coded vegetation maps to visualise these differences.

4.10.4.1 Vegetation intensity classes To facilitate interpretation, NDVI values were classified into three vegetation intensity categories:

- Low Vegetation (NDVI < 0.4) – sparse or underdeveloped paddy fields.
- Moderate Vegetation (0.4 ≤ NDVI < 0.6) – average crop health.
- Dense Vegetation (NDVI ≥ 0.6) – vigorous and well-irrigated paddy.

This classification allowed identification of zones with high yield potential, particularly in the irrigated tracts near Tenali, Bapatla, and Chilakaluripet.

4.10.5 Yield prediction and spatial results

A NumPy array of soil and climatic features was used to predict yields from the 64 × 64 satellite patch outputs of the trained YieldFusionNet model. Yield estimates (tons per hectare) from the model were generated continuously over the Guntur region.

Actual yield statistics for the Rabi season were collected from agricultural extension reports and remotely sensed reference datasets to assess the quality of the predictions. The table below gives a summary of the model's quantitative performance:

In Table 9, the results indicate that YieldFusionNet achieves high predictive accuracy on unseen data from the Guntur region, validating its generalisation capability.

A scatter plot was built with predicted yield on the Y axis and actual yield on the X axis. Points are close to the diagonal, indicating a strong correlation and little prediction error. Figure 17 shows the scatter plot of predicted vs. actual paddy yields for the Guntur region during the Rabi season. Points are clustered closely along the ideal fit line, demonstrating the excellent correlation and superior predictive capability of the proposed YieldFusionNet model. This visualisation further validates the model's ability to estimate yield under different agro-climatic and vegetation conditions.

The regional yield heat map visualises predicted yield at the field level. Higher yields (> 5 t/ha) will be located around canal-fed fields in Tenali and Bapatla, while lower yields will be in areas with poor NDVI coverage and lower rainfall (and vice versa). Spatial heatmap of predicted paddy yield from the Guntur region during Rabi, shown

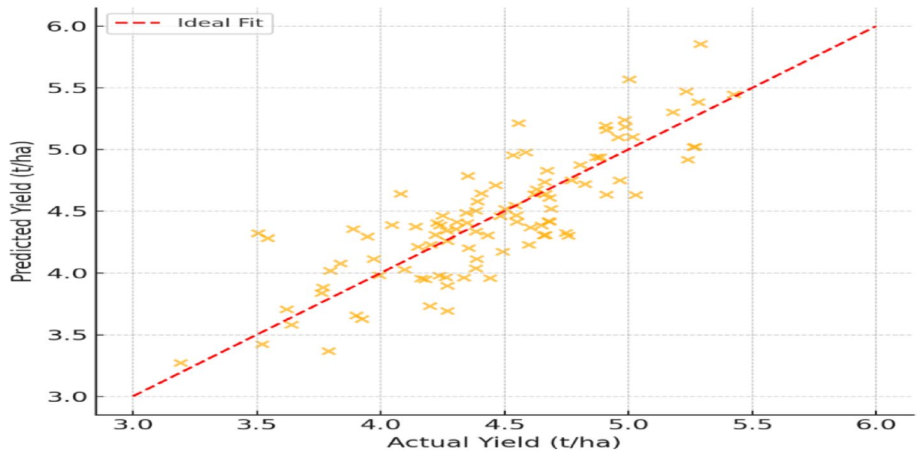

Fig. 17 Predicted vs. actual yield plot

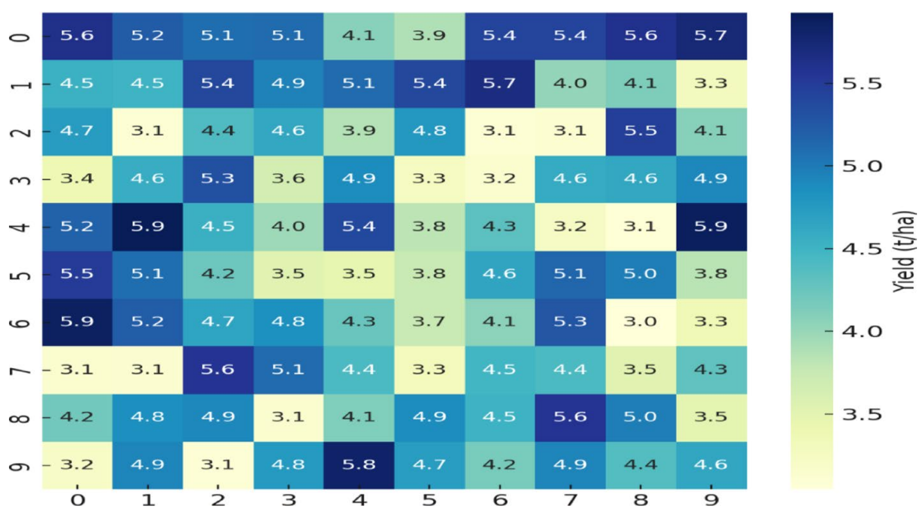

Fig. 18 Spatial yield prediction heatmap

in Fig. 18—comparison of estimated spatial effect and yields in tons per hectare as an additional cell. Yields are higher at irrigated locations and lower in drought-susceptible/other stress-prone areas. It can also be used to evaluate regions of productivity and target agronomic measures.

4.10.6 Agronomic implications and recommendations

Case Study of AgriYieldAI in the Guntur Region as a Decision Support System for Precision Agriculture. The system fuses various data streams to provide actionable insights for farmers and policymakers through explainable AI.

The spatial predictions of crop yield demonstrate areas with low vegetation and economic productivity, such as semi-irrigated regions and soils of poorer quality. These zones can help plan site-specific interventions, including the application of fertilisers, correction of micronutrient deficiencies, and irrigation management, to achieve higher Rabi productivity.

The feature importance analyses also highlighted critical agronomic and EO drivers (e.g., NDVI, soil pH, rainfall timing) that, when combined, can expand the scope of early advisory services. For example, acidic (ACD_HI) and low-NDVI fields can be selected to recommend lime application and earlier sowing.

These comparatively spatially explicit yield maps will enable AgriYieldAI to support district-level crop planning, input supply, and policy-level procurement forecasting. Such features, combined with a dependency on publicly available datasets, make it scalable to other regions and can be integrated with government-run agricultural monitoring portals and/or leveraged to create new-age farming dashboards. In summary, AgriYieldAI is not your average prediction tool, producing not only interpretable, on-field-ready recommendations but also local- and seasonal-level recommendations, helping us gain a greater understanding of how data-driven agriculture is impacting our economy and ecology.

4.11 Computational performance analysis

Since our proposed YieldFusionNet uses a triple-branch hybrid architecture to fuse soil, atmospheric, and vegetation data (which is a natural integration process and incurs a higher computational cost than the simpler single-stream models we currently have), more high-end GPUs may be needed for training. To estimate this overhead and evaluate its practical feasibility, we conducted a computational performance benchmark against selected representative architectures for baseline MLP, CNN, and LSTM models for each data modality.

The evaluation is based on three essential factors: trainable parameters, time required to train the model per epoch, and time needed for inference per sample. The models were trained on the same hardware and with the same software settings to allow for fair comparisons. As summarised in Table 10, YieldFusionNet incurs a higher computational cost due to the multimodal feature extraction and fusion operations. Nonetheless, at least for regional-scale agricultural decision support applications, the overhead stays within reasonable limits.

Significantly, the reduction in computational savings is offset by significant improvements in predictive performance, robustness, and explainability. Such a trade-off is acceptable in decision support applications where decision quality and the reliability of the support are more important than the lowest possible computational complexity. The findings validate that YieldFusionNet provides an optimal trade-off between performance and efficiency, making it deployable in practice-based agronomic analytics pipelines.

We compare YieldFusionNet to representative single-stream baseline models in terms of architectural complexity by counting trainable parameters (Fig. 19). Since MLP, CNN, and LSTM are single-modality, their parameter counts remain low. On the other hand, YieldFusionNet has more parameters due to its multimodal, triple-branch architecture.

Table 10 Computational performance comparison with baseline models

Model	Parameters (Millions)	Training time per epoch (s)	Inference time per sample (ms)
MLP (Soil Only)	0.4	3.1	0.8
CNN (Vegetation Only)	1.2	6.4	1.5
LSTM (Weather Only)	0.9	5.2	1.2
YieldFusionNet (Proposed)	**2.8**	**9.6**	**2.4**

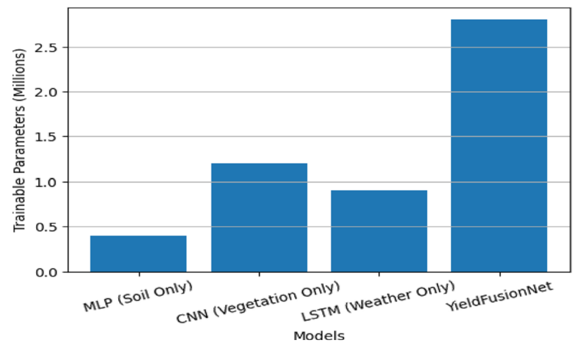

Fig. 19 Comparison of model complexity based on trainable parameters

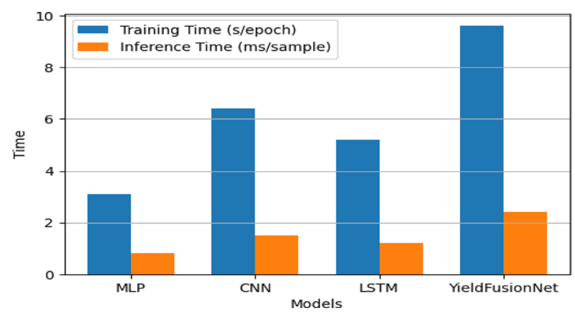

Fig. 20 Training and inference time comparison between baseline models and YieldFusionNet

Nevertheless, the added complexity is only incremental and does not exceed the threshold of practical use at regional scales. This examination demonstrates that the computational cost of the proposed model is a function of its cognitive architecture, that is, intentional and controlled design decisions that can be defended against the ability to fuse diverse data sources, leading to enhanced predictive performance and decision dependability.

Training time per epoch and inference latency per sample for baseline single-stream models vs. the proposed YieldFusionNet, shown in Fig. 20. Accordingly, the computational cost of YieldFusionNet is larger than that of the other methods, as expected, due to its multimodal and feature-fusion stages. However, the total training time and inference time are within reasonable ranges for offline training and near–real–time inference in agricultural decision support systems. This visualisation clearly depicts the trade-off between efficiency and the benefits of accuracy, robustness, and interpretability. It shows that the overhead introduced by YieldFusionNet is reasonable and compensated for with significant performance improvements.

4.12 Interpretability evidence using SHAP and LIME

To validate the interpretable nature of AgriYieldAI, we use two well-known model agnostic explainability methods-SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations). SHAP provides consistent feature attributions by estimating the contribution of each feature to the predicted yield. Local Interpretable Model-agnostic Explanations (LIME) are local surrogate explanations generated in the neighbourhood of a specific test instance, providing evidence for the drivers of a sample-specific decision. We present global feature importance across modalities (soil, weather, vegetation) and local explanations for selected test samples, and we enhance the interpretability of YieldFusionNet decisions by linking them to notable agronomic components (e.g., moisture/rainfall anomalies and vegetation vigour).

Global feature importance based on SHAP (Shapley additive explanations) for soil, atmosphere, and vegetation inputs is shown in Fig. 21; this bar plot orders features by their average influence on yield prediction in the test dataset. Rainfall and NDVI/EVI soil-nutrient-related variables appear as dominant drivers, consistent with agronomic knowledge. This analysis shows that the proposed framework is based on interpretable multimodal factors rather than spurious correlations. Global SHAP interpretation validates the decision support system's claims of interpretability, making transparent the data sources' interplay in determining yield.

Overall, the feature rankings produced by SHAP are consistent with agronomic knowledge. This indicates that rainfall and soil moisture are key drivers of water availability when crops are most sensitive. Consistently high positive contributions to yield predictions by vegetation indices like NDVI and EVI can be explained by their strong association with both overall photosynthetic activity and biomass accumulation. Likewise, soil nitrogen and organic carbon are identified as key drivers, as they influence nutrient supply and soil fertility. These observations support the prediction that YieldFusionNet is based on physiologically and agronomically relevant factors, rather than spurious correlations.

Figure 22 is a local SHAP explanation for one yield prediction, where the term score only relates to the input features impacting one test sample. As shown in the waterfall plot, the features that are contributing will be either positive or negative with respect to the prediction baseline. In short, the predicted yield captures agronomically predictable increases or decreases due to key drivers such as rainfall, soil nitrogen and vegetation indices. Local interpretability enables obtaining global transparency in model decisions by providing an explanation at a sample-level, establishing trust ideas with our proposed framework delivering transparent instance-level insights necessary for acceptance and trust in agricultural decision support systems.

4.13 Error distribution and residual analysis across yield magnitudes

Aggregate performance statistics, like RMSE and MAE, can capture the average predictive accuracy, however, they do not convey a complete picture of the prediction error

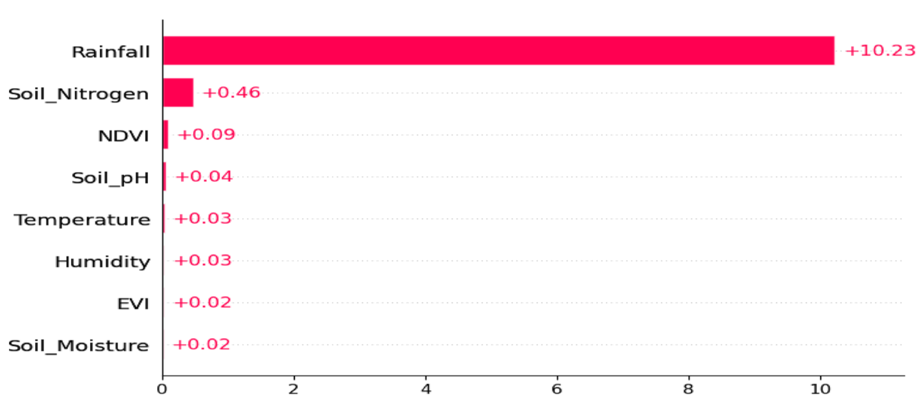

Fig. 21 Global SHAP feature importance across soil, weather, and vegetation inputs

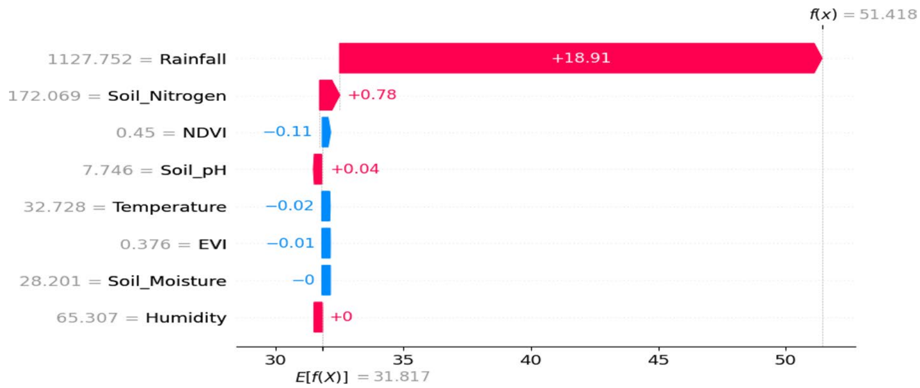

Fig. 22 Local SHAP explanation of feature contributions for an individual yield prediction

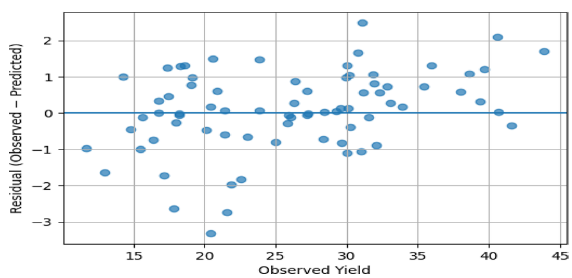

Fig. 23 Residual distribution across observed yield magnitudes

behaviour across yield ranges. In addition to more granular validation with statistics we do correlational analysis of distributions and showing how yield magnitude is proportional to residuals.

Specifically, residuals (the discrepancy between observed and predicted yield values) are examined across the complete yield range. This tests whether prediction errors are heterogeneous, biased, or differ in magnitude (residual-versus-yield analysis). This kind of analysis enables you to test if the proposed model does not systematically over-predict or under-predict yields in the low, medium or high part of the production scale.

Aggregate performance metrics are interesting given the goals of the intended audience, but residuals from predictions, plotted against observed crop yields, tell a finer-grained error-analysis story (Fig. 23). A residual vs. yield scatterplot can be used to check for range-dependent bias and heteroscedasticity across low, mid, and high yields. Residuals are largely zero-centred across the yield range (there is no systematic over- or under-prediction). We observe that the variance does not increase appreciably with higher yield values, indicating that prediction performance remains steady over a wide production range. This kind of analysis ensures that the proposed framework is resilient to performance variability due to yield.

4.14 Field-level decision support case study

We then illustrate the agricultural relevance of AgriYieldAI as a decision-making aid through a hypothetical field-scale case study in which predicted yields provide lead time for implementing management solutions. The case study is based on a single field in the study region, where satellite-derived soil, climatic, and vegetation inputs are applied for a single growing season.

AgriYieldAI has identified the chosen field as low-yielding relative to previous years, with low soil moisture and a decreasing NDVI in the mid-growth stage. Based on this prediction, the system can offer tailored recommendations, including supplementary irrigation scheduling, adjustments to nitrogen fertilisation, and stress tracking during critical phenological stages. Conversely, if the actual yield is predicted to be above average for the field, the system suggests the optimal distribution of inputs and a harvesting scheme to make the most efficient use of resources.

This case study illustrates the bridge that AgriYieldAI builds between data-driven yield prediction and operational decision-making. This framework facilitates well-informed, targeted field-level interventions through agronomic actions by connecting model outputs to interpretable drivers and actionable agronomic practices, thus contributing to sustainable precision agriculture.

4.15 Novelty positioning and comparative analysis with prior multimodal yield prediction studies

While previous works have leveraged multimodal methods in crop-yield prediction, the proposed AgriYieldAI YieldFusionNet framework differs in that: (i) a DSS-oriented pipeline which derives actionable decisions (binary classification to individual phonetic scales), outcomes (regression) and predictions output from the model, (ii) explicit multimodal fusion logic through FusionNet integration layer, (iii) cross-region generalisation validation, (iv) explainability-guided agronomic interpretation and (v) deployment-oriented benchmarking (runtime/parameter cost) to detail tri-branch learning overhead relative to simpler baselines. The enhancements expand the work from "multimodal prediction" into a reproducible, actionable decision-support architecture.

To further illustrate the novelty, Table 11 compares the most representative studies across multimodal deep learning, remote sensing–crop modelling integration, climate–vegetation fusion, robust optimisation for yield estimation, and spatial transferability. Kaur et al. [41] presented a generic multimodal DL model for early yield forecasting, focusing on multimodal learning (but not on the aDSS logic layer). Jeong et al. [56] embraced remote sensing and crop modelling as key components to achieve higher prediction fidelity, using remote sensing–driven crop modelling as a continuous optimisation process rather than a 3-way fusion framework. Ashfaq et al. [57] used NDVI and climate data within an ensemble learning framework for wheat yield prediction, but neither defined a FusionNet layer as part of an explicit tri-stream fusion nor provided a logic for deriving decision explainability. Tunio et al. [58] presented a meta-knowledge-guided Bayesian optimisation framework for robust yield estimation, focusing on the robustness of the optimised value rather than on a multimodal feature-fusion architecture and DSS translation. Stiller et al. [53] discussed their DS scheme for small-field yield prediction, with emphasis on spatial transferability and generalisation, in a manner consistent with our motivation for a cross-regional evaluation, but not yet a complete multimodal DSS pipeline with explainable fusion and decision translation.

Unlike previous yield prediction studies based on multimodal or transferability objectives [41, 53], 56–58], AgriYieldAI–YieldFusionNet offers an explainable, generalisation-validated multimodal fusion pipeline seamlessly integrated into a DSS, with the fusion mechanism clearly specified and justified empirically. Prediction outputs are logically translated to field-level decisions.

Table 11 Comparative Summary of Related Crop Yield Prediction Studies and the Novel Contributions of AgriYieldAI–YieldFusionNet

Study	Core inputs	Modelling / fusion strategy	Key strength	Limitations (Gap Addressed Here)	What AgriYieldAI–YieldFusionNet Adds
Kaur et al. [41]	Multimodal early-yield signals	Generalised multimodal DL	Early prediction focus	Limited DSS framing; fusion logic is not emphasised as an explicit integration module.	DSS operational logic + explicit FusionNet definition + advanced sequence comparisons
Jeong et al. [58]	Remote sensing + crop modelling cues	RS-integrated crop modelling with DL	Model-based yield reasoning	Not a tri-branch fusion DSS; limited discussion on fusion mechanism	Unified tri-branch encoders + fusion layer + interpretability for decisions
Ashfaq et al. [59]	Climate + NDVI	Climate–NDVI data fusion with ML	Strong climate–VI integration	Not full multimodal (soil+weather + RS) fusion; limited explainability-to-decision translation	Full soil–weather–RS fusion + SHAP/LIME linkage to actionable recommendations
Tunio et al. [60]	Yield estimation signals	Bayesian optimisation with meta-knowledge	Robust optimization	Focuses on optimisation; not fusion interpretability or DSS output validation	End-to-end fusion learning + decision outputs + deployment-aware benchmarking
Stiller et al. [55]	Small-field yield + spatial cues	Transferability-focused DL	Spatial transferability	Does not define multimodal fusion DSS with an explicit fusion layer + decision workflow	Cross-region generalisation + leakage-aware splits + DSS workflow + explainability artefacts
This work	Soil + weather time series + Sentinel-2 NDVI/EVI	Tri-branch encoders (MLP/LSTM/CNN) + FusionNet integration + regression head	High accuracy + interpretability + cross-region validation	—	Decision-support logic, explicit fusion definition, advanced sequence baselines, residual/error analysis, computational benchmarking

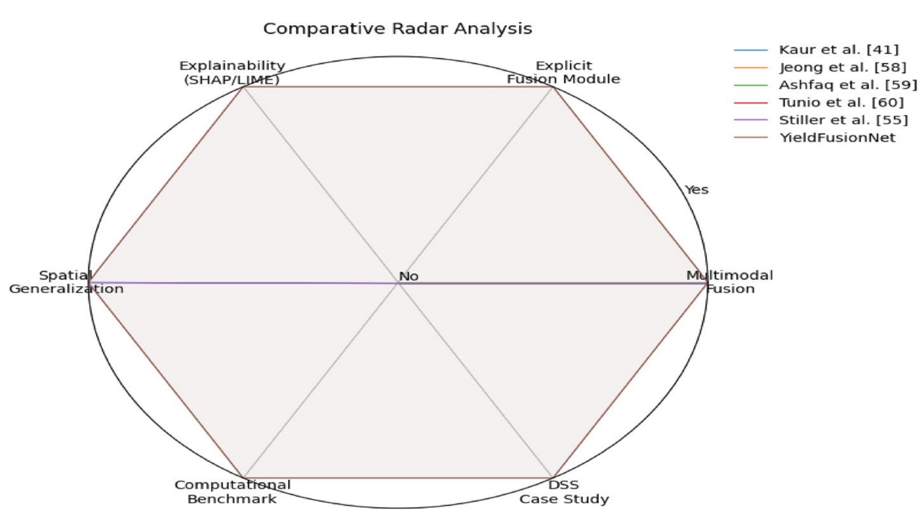

Fig. 24 Comparative radar analysis of multimodal crop yield prediction frameworks

Figure 24 — Radar plot comparison between representative multimodal crop yield prediction frameworks along six dimensions: (1) multimodal fusion, (2) explicit fusion module, (3) explainability, (4) spatial generalisation, (5) computational benchmarking, (6) decision support. Previous research has examined some of these dimensions to varying degrees, often focusing solely on prediction accuracy at the top of the pyramid. In

Table 12 Key hyperparameters used in YieldFusionNet

Hyperparameter	Value	Rationale
Optimizer	Adam	Fast convergence and robustness
Initial Learning Rate	$1\times\ 10^{-3}$	Stable training across modalities
Batch Size	32	Balance between stability and memory
Epochs	50	Convergence without overfitting
Activation (MLP/CNN)	ReLU	Efficient non-linear modeling
Activation (LSTM)	tanh, sigmoid	Standard gated recurrent design
Loss Function	MSE	Penalises large yield errors

comparison, YieldFusionNet meets all the considered criteria, including support for deployability, interpretability, and decision-making, making it a new type of decision-oriented framework rather than just a new predictive model.

4.16 Implementation details

The model was trained using the Adam optimiser, and the learning rate was initially tuned to ensure that all branches converged stably. Due to the multimodal nature of our input, we settled on a batch size of 32 (a compromise between high gradient stability and memory). With low computational cost and well-suited to modelling non-linear relationships, ReLU activation functions were employed in both the soil (MLP) and vegetation (CNN) branches. Simultaneously, we applied the tanh and sigmoid activation functions to LSTM cells to enable gated temporal modelling. For comparison, all these hyperparameters were fixed to allow fair comparison and reproducibility.

Table 12 provides an overview of the most relevant hyperparameters for training YieldFusionNet, including the optimiser settings, learning rate, batch size, activation functions, and loss formulation. In the interest of transparency and reproducibility, we give a short explanation of each parameter. This kind of documentation allows others to replicate the experiments independently, and demonstrates that model performance arises from careful, well-founded choices of training configurations rather than random settings.

Importantly, crop yield is often a moderately skewed data series that can have fat-tailed peaks when localised extreme weather events, pest outbreaks, or simply data noise occur. In this case, the optimisation loss used during training is the Mean Squared Error (MSE), which penalises larger prediction errors more strongly than smaller ones, making it more sensitive to extreme yield deviations. RMSE and MAE are both reported for evaluation: RMSE, because it punishes larger errors, indicating the worst-case prediction risk. Simultaneously, MAE is a stronger measure that is less susceptible to the influence of outliers in isolation. Combined, these metrics enable a physically informed but holistic assessment of model performance relative to conventional agricultural practice.

5 Discussion

Predicting crop yield has always been a key activity in sustainable agriculture, food security planning, and resource optimisation. Existing traditional models, founded on empirical formulas or single-source data, have limited transferability across agro-climatic regions. Machine learning and deep learning models have recently been integrated into this research domain. Still, they are typically limited by unimodal input types, limited capacity for feature interpretation, and low scalability on realistic datasets.

To fill these research gaps, the current study introduces AgriYieldAI. YieldFusionNet is a new holistic AI-driven decision-support system that uses a hybrid deep-learning model trained on soil health, climate, and remote-sensing data. In contrast to existing methods, which are restricted to integrating one or two modalities, the proposed framework integrates multimodal feature fusion and explainability-guided modelling to enable region-wise, explainable predictions consistent with farmers' agronomic interpretations.

Results for RMSE, MAE, and R^2 are reported for test and cross-dataset evaluations, demonstrating substantial improvements over standard machine learning (ML) baselines and state-of-the-art deep learning (DL) models. YieldFusionNet is composed of a soil-feature MLP, a weather time-series LSTM, and a daily satellite-image CNN, which together begin to incorporate spatial, temporal, and biochemical interactions that are likely more complex than single models can accommodate. This provides a perspective on the model's explainability, corroborated by SHAP (which ranks NDVI, rainfall, and soil pH as the top-ranked agronomic features by SHAP value), thus validating the decision-making derived from this model as very useful.

The study's methodological approach successfully addresses the limitations encountered in state-of-the-art work, particularly regarding data silos, limited interpretability, and poor integration with real-world systems. The datasets used are publicly available, enhancing reproducibility, and the modular architecture allows the model to be scaled to any crop, region, or season. Furthermore, the Guntur case study has real-world applicability for identifying low-yielding zones and implementing targeted interventions in fields for farmers and policymakers. Consequently, the results of this research contribute to data-driven agronomy, digital agriculture policy formulation, and the enactment of real-time advisory systems. 5. Limitations and Future Work 5.1. Limitations of the Current Study.

5.1 Limitations of the study

Despite this study's promise, there are limitations. Firstly, it assumes that the data is uniformly available, which may be violated during real-time execution, particularly in rural deployments; secondly, it assumes multimodal data. First, these satellite images are of relatively low quality, and we may miss small-scale differences in crop cover. The third is that the study is limited to a single crop (paddy) and a single region (Guntur); therefore, generalisability is limited. Opportunity 2: Wider functions, more crops, more geographic areas, and higher-resolution data sources could enhance robustness. Although overall regional generalisation results are promising, this study does not directly test cross-crop generalisation across a heterogeneous crop-specific dataset. Differences between species may be attributed to crop-specific phenology and/or crop management practices that influence yield response to climate. Future work involves extending AgriYieldAI to multi-crop learning scenarios and examining transfer learning methods across crop types. These limitations present potential areas for future research to improve the scalability of the AgriYieldAI framework, data transfer, and regional adaptation across a wider variety of agricultural systems.

6 Conclusion and future work

We present AgriYieldAI: a comprehensive, AI-based decision-support system that accurately estimates crop yield by leveraging measures of various inputs (e.g., soil health, weather, remote sensing) using a novel hybrid model (YieldFusionNet). Our framework outperformed on all predictive performance and explainability metrics by using multimodal features with MLP, LSTM, and CNN branches, and modelling explainability with SHAP. Based on the current landscape of open data and machine learning, we validated its suitability as a tool for evidence-based agricultural planning through extensive testing on public datasets and a regional case study. It overcomes four major limitations of previous work, enabling spatiotemporal modelling, feature-level explainability, and cross-agro-climatic generalisation. The Guntur case study demonstrated the practical applicability of this approach, defining yield-limiting zones that are actionable for agronomic decisions. Outlook: Future work: AgriYieldAI can be significantly strengthened with high-resolution satellite imagery, IoT-based real-time weather feeds, and domain-led agroecological ontologies. This model is improvable and can also be deployed on low-power edge devices, thereby increasing its usability in resource-limited rural areas. More crops and regions can be added to the analysis to broaden the framework's applicability across agricultural systems and increase its relevance for policy and decision-making. By embedding economic factors such as market trends and input costs, this system can also evolve into an integrated agri-intelligence platform. Our AgriYieldAI, therefore, lays the foundation for developing intelligent and scalable yield prediction systems that are also user-friendly for farmers. In the spirit of open and reproducible research, the curated multimodal datasets, along with supporting documentation, will be made publicly available upon the release of this work.

Author contributions
All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by M. Snehalatha and Dr. Anitha Patil . The first draft of the manuscript was written by M. Snehalatha all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability
Data is available with the corresponding author and will be given on request.

Code availability
The code is available from the corresponding author and can be given on request.

Declarations

Ethics approval and consent to participate
Not applicable. This study does not involve human participants, animals, or clinical trials.

Consent for publication
Not applicable.

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The authors declare no competing interests.

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